

Joint identification of monetary policy, Fed-response-to-news, and risk premium shocks*

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Abstract

YYYYYYYYFOMC announcement surprises mix four shocks: monetary policy, central bank information, the Fed's response to public news, and the risk premium. Existing methods identify three at a time. I identify all four jointly from four asset-price surprises in a narrow window around each announcement, using sign and magnitude restrictions from an affine asset pricing model. All four shocks are present in the window: the two-year yield surprise is shared by the policy, information and response-to-news shocks, the equity surprise is mostly the policy shock, and the ten-year yield surprise is mostly the risk premium shock, at more than 60% of its variance. Outside the announcement window, it lowers the ten-year yield through the term premium, with no detectable response of the yield's expectations component, credit spreads, or risk appetite: it is a term premium shock. Omitting the risk premium shock leaves the policy shock and the monthly macroeconomic responses mostly unchanged, but about a tenth of what the three-shock system labels central bank information becomes risk premium variation. Omitting the Fed's response to news instead makes a fifth of what that system labels monetary policy the Fed's response to news, and underestimates the effects of policy on output and prices. What moves long-term rates at announcements is mainly the compensation for bearing interest-rate risk, not news about policy.

Keywords: monetary policy shocks, risk premium, central bank information effect, Fed response to news, stock-bond co-movement

JEL Classification: E52, E58

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1 Introduction

Almost every aggregate variable in macroeconomics is endogenous: policy responds to the state of the economy and the state of the economy responds to policy. This makes estimation of causal effects of monetary policy challenging, and requires a method that can identify policy moves orthogonal to the state of the economy, i.e. structural policy shocks. Among the methods proposed in the literature, high-frequency identification is one of the most potent. This approach measures the change in price for some financial asset in a tight window around the announcement (monetary policy surprise, MPS) and uses it as an instrument for the causal effects of monetary policy (Kuttner, 2001; Gurkaynak et al., 2005; Nakamura and Steinsson, 2018). The idea of the approach is based on the assumption that asset prices move within a tight (usually 30-minute) window around the announcement only due to its unanticipated content, thus representing the effect of the unexpected (not driven by the state of the economy) part of the policy change.

However, despite being unanticipated, MPS is not exogenous: within the announcement window, the market surprise combines several distinct structural policy innovations. In simple words, the policy surprise is a combined reaction of the market to different policy news present in a single announcement. Suppose for simplicity that the Fed sets the policy rate (r_t) in response to growth factor g_t , $r_t = a \mathbb{E}_t^{cb}[g_t] + \varepsilon_t^{MP}$, where ε_t^{MP} is a deviation from the systematic part of the policy rule. Suppose the market's pre-announcement expectation of the rate is $\mathbb{E}_{t-}^m[r_t] = a^M \mathbb{E}_{t-}^m[g_t]$, where a is the true growth-response coefficient and a^M the market's belief about it. The high-frequency monetary policy surprise, $\text{MPS}_t \equiv r_t - \mathbb{E}_{t-}^m[r_t]$, then decomposes into three pieces:

$$\text{MPS}_t = \underbrace{\varepsilon_t^{MP}}_{\text{MP}} + \underbrace{a(\mathbb{E}_t^{cb}[g_t] - \mathbb{E}_{t-}^m[g_t])}_{\text{CBI}} + \underbrace{(a - a^M) \mathbb{E}_{t-}^m[g_t]}_{\text{FRN}}. \quad (1)$$

The first term is the *monetary policy* (MP) component, an exogenous deviation ε_t^{MP} from the systematic policy rule (Ramey, 2016). The second is a *central bank information* (CBI) component: the policy surprise part due to the Fed revealing a different outlook from the market's pre-announcement expectations (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020). The third is a *Fed response to news* (FRN) component: the market's belief a^M about the Fed's reaction coefficient differs from the true a , so the term $(a - a^M) \mathbb{E}_{t-}^m[g_t]$ is a function of information the market already had, which makes the surprise predictable *ex post* from public pre-announcement information (Bauer and Swanson, 2023a,b; Swanson, 2024).

A fourth shock moves asset prices in the window without generating a policy rate sur-

prise: a *risk premium* (RP) shock, a change in the compensation investors require for bearing risk, driven by shifts in risk appetite or in perceived risk (Bauer et al., 2023; Kroencke et al., 2021; Boehm and Kroner, 2024). Equation (1) has no risk premium component; however most asset prices do, and Cieslak and Schrimpf (2019) and Cieslak and Pang (2021) show that the premium moves within the announcement window. The premium is small at the maturities the policy surprise is built from, which is why identifications based on the rate surprise alone usually ignore it (Cieslak and Schrimpf, 2019; Miranda-Agrippino and Williams, 2026). It is neither small nor constant in long yields and in equities. Bernanke and Kuttner (2005) decompose the equity response to the funds rate surprise and attribute most of it to news about expected excess returns rather than about dividends or real rates. Hanson and Stein (2015) find that policy announcements move distant real forward rates and attribute the movement to term premia. Cieslak and Schrimpf (2019) and Cieslak and Pang (2021) show that the joint window behavior of stocks and short- and long-maturity yields contains a component with the covariance signature of risk premium news, and Kroencke et al. (2021) and Boehm and Kroner (2024) isolate announcement-window components orthogonal to the rate surprise itself. Investors are compensated simply for holding risk through a scheduled announcement: over 1994–2011, the twenty-four hours before FOMC announcements accounted for more than 80 % of the U.S. equity premium (Lucca and Moench, 2015).

Risk premia move at announcements for two reasons. The first is the rate news itself: a rate surprise moves risk aversion, and premia move with it, as in the habit model of Pflueger and Rinaldi (2022). That response is part of the transmission of the MP, CBI, and FRN shocks defined above; it is the risk-taking channel of monetary policy (Borio and Zhu, 2012). The second is premium movement that no rate news causes, and that is the RP shock. Announcements can produce it in at least two ways. First, a scheduled announcement resolves uncertainty: policy uncertainty falls on average when the statement is released (Bauer et al., 2022), and the resolution alone changes what investors require for bearing risk, even when the announced path of the policy rate is identical to the expected one (Ai and Bansal, 2018). A related channel: a statement that shifts the balance of risks changes perceived risk, and premia move with it. Second, the statement and the press conference contain more than the path. Cieslak and Schrimpf (2019) find that non-monetary news accounts for more than half of the content of the communications that explain policy decisions across the central banks they study, and they read its risk premium part as news that moves investors’ risk appetite or sentiment separately from fundamentals. News about asset purchases moves long yields without moving the expected short rate (Swanson, 2024), and it also changes the scarcity of safe collateral, which moves the compensation investors require for bearing risk (Caballero et al., 2017).

Each component of (1) is driven by a single underlying structural shock, ε_t^{MP} , ε_t^{CBI} , ε_t^{FRN} , which together with the risk premium shock ε_t^{RP} gives four mutually orthogonal shocks present in the policy announcement window. Separating them matters because each answers a different question: MP shocks instrument the transmission of policy rates to output, prices, and financial conditions; CBI shocks measure what markets learn about the economy from the Fed; FRN shocks isolate the predictable part of the surprise, which must be removed for the surprise to be a valid instrument (Bauer and Swanson, 2023a,b); and RP shocks measure what exogenous premium shocks do to markets and activity.

No existing identification accommodates all four. The Fed’s response to news and the risk premium have been studied in two strands of literature that developed separately. Jarociński and Karadi (2026) (hereafter JK) identify MP, CBI, and FRN shocks using sign and magnitude restrictions on three high-frequency observables: the Bauer and Swanson (2023b) fitted value and residual, the components of the policy surprise that are respectively predictable and unpredictable from pre-announcement information, and the S&P500 surprise. JK assume away the risk-premium channel. Cieslak and Schrimpf (2019) (hereafter CS) identify MP, CBI, and RP shocks from the joint window behavior of equities and short- and long-maturity Treasury yields, and assume away the FRN channel instead. The CS risk premium shock captures windows in which stocks and yields fall together, with long yields falling by more than short ones: investors shift from risky assets toward safer Treasuries, pushing bond prices up and yields down, especially at the long end. Each scheme rules out precisely the shock the other treats as important.

When a structural shock is omitted, its variation does not disappear. It is absorbed by the shocks that remain, which then become contaminated. For example, the contamination from missing CBI and/or FRN, when not identified together with MP shock, has been analyzed and documented (Jarociński and Karadi, 2020, 2026; Bauer and Swanson, 2023a). There is no evidence about the contamination that involves the risk premium shock, in either direction, and there is an a priori reason to look for it. Bauer and Swanson (2023a) show that the FRN component accounts for a substantial share of the variance of the policy surprise, though the size of the channel remains debated (Miranda-Agrippino and Williams, 2026). The CS three-shock system contains no observable that isolates that component, so its variation must be absorbed by some combination of the MP, CBI, and RP shocks. Likewise, the risk premium shock is a large component of announcement surprises (Cieslak and Schrimpf, 2019) and is likely absorbed by some combination of the shocks considered in JK. A related conjecture of Swanson (2025) is that part of what the literature identifies as the central bank information effect may in fact be risk-premium variation.

Neither system can say where the missing shock’s variation ends up, or how large it is.

Answering that requires a system in which all four shocks are identified together. This paper provides the first joint identification of MP, CBI, FRN, and RP at FOMC announcements. I propose a unified sign- and magnitude-restriction framework, henceforth JKCS, that combines the Jarocinski–Karadi and Cieslak–Schrimpf systems, so that each strand’s shock is identified in the presence of the other’s. JKCS retains the Bauer–Swanson decomposition of the policy surprise into a residual (RES) and a fitted value (FIT), which is what separates MP shock from FRN; keeps the high-frequency S&P500 surprise, which separates the central bank information shock (Jarociński and Karadi, 2020); and adds the ten-year Treasury yield as a fourth observable, which supplies the maturity dimension required to separate the risk premium shock from the other shocks that move stocks and long yields together. The resulting 4×4 system is set-identified by sign and magnitude restrictions of the kind used in JK and CS individually, supplied from a simple affine asset pricing model, together with an external correlation restriction that disciplines the risk premium shock with the daily risk-appetite index of Bauer et al. (2023).

Three results follow, and together they reconcile the two strands. First, the risk premium shock is a long-end phenomenon. It accounts for about 60–70% of the variance of the ten-year yield surprise within announcement windows and for almost none of the two-year yield or equity surprises, and it is the largest single contributor to the long end in every one of seven configurations of the identification. An announcement moves long rates mainly by changing the compensation investors require, not by changing the expected path of policy. The risk premium shock label holds outside the window: in daily local projections the shock lowers the ten-year yield through the term premium of Adrian et al. (2013), with no reaction from the expectations component, and at the monthly frequency it is a financial repricing with little macroeconomic effects. The shock survives controlling for the other three shocks, and it is not created by its own restrictions: with every identifying restriction on the fourth column dropped, the shock that emerges still carries the risk premium signature.

Second, each omission contaminates a different shock. Omitting the response-to-news shock, as CS does, leaves the risk premium shock intact: decomposed into the four JKCS shocks, the CS risk premium shock is 98.8% the jointly identified one. This is different for the policy shock, about 79% of which is the jointly identified policy shock and 20% corresponds to the response-to-news shock, the direction Section 2.6 and Bauer and Swanson (2023a) predict. Omitting the risk premium shock, as JK does, leaves the policy and news shocks close to intact; the cost falls on the information shock, 20.7% of which is something else, mostly risk premium (8.9%) and response to news. The 8.9% answers the conjecture of Swanson (2025): part of the measured information effect is in fact risk-premium variation, but this part is small and does not change the identified responses of financial and macro

variables to CBI shock substantially.

Third, the macroeconomic implications of omission for measured responses are likely limited. At both daily and monthly frequencies the responses of the shocks the systems share stay inside each other’s bands, and for the JK shocks the responses are almost identical for all macro and financial variables considered. What differences remain concentrate in the policy (MP) shock: in daily local projections the CS policy shock overstates the credit and risk-appetite responses of the policy shock identified here by a quarter to two fifths, and in the monthly VAR the policy column is the one place where the CS and JKCS median responses separate from each other, with the larger S&P500 and price-level responses on the JKCS side. For applied work the implication is asymmetric: the JK policy shock needs no correction for the omitted premium, while a policy shock identified without a response-to-news shock mixes monetary policy with the Fed’s predictable response to news.

Section 2 develops an affine term-structure model of the announcement window that derives the sign restrictions used to identify the shocks. Section 3 introduces the three systems and their restrictions, and states the two comparability choices that make the three sets of shocks measurable on the same windows. Section 4 describes the data. Section 5 reports what each shock explains within the announcement window, on average and announcement by announcement, and how the decomposition changes relative to JK and CS. Section 6.1 asks whether the four shocks move financial variables that played no part in identifying them in the directions their labels predict. Section 6.2 follows with the reaction of macro variables at the monthly frequency. Section 7 draws the implications for applied work, and Section 8 concludes.

2 An illustrative model of the four shocks

The identification in Section 3 rests on sign and magnitude restrictions on four announcement-window observables. This section derives those restrictions in a simple asset-pricing model. I reformulate the Cieslak and Schrimpf (2019) affine model and introduce a Fed-response-to-news shock into it, derive the window responses of bond yields and stock prices to each of the four shocks.

2.1 State vector and dynamics

Suppose that the economy is described by the state:

$$F_t = (g_t, i_t, x_t)',$$

where g_t is growth factor, i_t the short rate, and x_t the price-of-risk factor. The state follows a VAR(1),

$$F_{t+1} = \Phi F_t + \Sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I_3) \text{ i.i.d.}, \quad \varepsilon_{t+1} = (\varepsilon_{t+1}^g, \varepsilon_{t+1}^i, \varepsilon_{t+1}^x)'$$

where all state variables are expressed as deviations from unconditional means; the constant vector is suppressed and:

$$\Phi = \begin{pmatrix} \phi_g & 0 & 0 \\ a\phi_g & \phi_i & 0 \\ 0 & 0 & \phi_x \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_g & 0 & 0 \\ a\sigma_g & \sigma_i & 0 \\ 0 & 0 & \sigma_x \end{pmatrix}, \quad \phi_g, \phi_i, \phi_x \in (0, 1).$$

The formulation above follows directly when I assume that the growth and risk factors follow AR(1) processes:

$$\begin{aligned} g_{t+1} &= \phi_g g_t + \sigma_g \varepsilon_{t+1}^g \\ x_{t+1} &= \phi_x x_t + \sigma_x \varepsilon_{t+1}^x, \end{aligned}$$

while the policy rate i_{t+1} is a persistent process that responds to the same-period growth g_{t+1} with a fixed coefficient a (the *true* growth-response coefficient; the market's belief a^M about it is introduced in Section 2.3). Written explicitly, the policy rate is:

$$\begin{aligned} i_{t+1} &= \phi_i i_t + a g_{t+1} + \sigma_i \varepsilon_{t+1}^i = \phi_i i_t + a(\phi_g g_t + \sigma_g \varepsilon_{t+1}^g) + \sigma_i \varepsilon_{t+1}^i \\ &= \underbrace{a\phi_g g_t}_{\text{dynamic resp.}} + \phi_i i_t + \underbrace{a\sigma_g \varepsilon_{t+1}^g}_{\text{contemp. resp.}} + \underbrace{\sigma_i \varepsilon_{t+1}^i}_{\text{MP}}. \end{aligned}$$

The model makes two simplifications. First, growth factor does not respond to policy: the g -row of Φ and Σ contains no i (Cieslak and Schrimpf, 2019). This simplification allows to separate news about cash flows from news about discounting in the announcement window. Allowing moderate feedback, $g_{t+1} = \phi_g g_t - \theta i_t + \sigma_g \varepsilon_{t+1}^g$ with $\theta > 0$, would not change the qualitative results (Cieslak and Pang, 2021). Second, the policy rate responds to growth factor and not to expected inflation, so g_t could be read as the single state the Fed reacts to rather than as real (GDP) growth narrowly.

2.2 SDF and pricing

Having described the economy, we now turn to asset prices. The identification in Section 3 relies on the signs of the announcement-window responses of asset prices to each shock, so

what the model has to deliver is the mapping from the shocks of Section 2.1 to bond yields and stock prices. This subsection derives that mapping.

I start from the general pricing principle that for an asset with price 1 today, the expected discounted payoff must equal the price, $\mathbb{E}_t[M_{t+1}R_{t+1}] = 1$, where R_{t+1} is a gross return and M_{t+1} is the stochastic discount factor (SDF). In a consumption-based model the SDF is the intertemporal marginal rate of substitution, $M_{t+1} = \beta u'(C_{t+1})/u'(C_t)$, high in states where consumption is scarce: an asset paying badly in exactly those states must compensate its holder, while one whose payoff is uncorrelated with M_{t+1} earns only the risk-free rate, however volatile it is.¹

Assume the log SDF, $m_{t+1} = \ln(M_{t+1})$, is linear in next period's shocks,

$$m_{t+1} = \mu_t - \lambda_t' \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I_3),$$

where μ_t (a scalar) and λ_t (a 3×1 vector) are known at date t . The one-period safe bond pins down $\mu_t = -i_t - \frac{1}{2}\lambda_t' \lambda_t$,² so

$$m_{t+1} = -i_t - \frac{1}{2}\lambda_t' \lambda_t - \lambda_t' \varepsilon_{t+1}.$$

The vector λ_t is the price of risk. The SDF innovation is $m_{t+1} - \mathbb{E}_t[m_{t+1}] = -\lambda_t' \varepsilon_{t+1}$, so a shock is compensated, that is *priced*, if and only if its entry in λ_t is nonzero; exposure to a shock with a zero entry is orthogonal to m and earns nothing beyond risk free rate. I specify

$$\lambda_t = (\lambda_{gx} x_t, 0, 0)', \quad \lambda_{gx} > 0,$$

which says two things: investors are compensated for growth risk alone, and the price they charge for it moves with x_t .³

The first restriction follows the usual interpretation of the SDF. Though it is specified here rather than derived from preferences, following Cieslak and Schrimpf (2019) and the affine no-arbitrage literature, m can be interpreted as the intertemporal marginal rate of substitution and so moves with news about the consumption stream (Cochrane, 2009). There

¹Expanding $1 = \mathbb{E}_t[M_{t+1}R_{t+1}] = \mathbb{E}_t[M_{t+1}]\mathbb{E}_t[R_{t+1}] + \text{Cov}_t(M_{t+1}, R_{t+1})$ and using $\mathbb{E}_t[M_{t+1}] = 1/R_t^f$, where R_t^f is the gross risk-free rate, gives $\mathbb{E}_t[R_{t+1}] - R_t^f = -R_t^f \text{Cov}_t(M_{t+1}, R_{t+1})$: the premium is positive iff the return covaries negatively with the SDF, and zero if they are uncorrelated.

²The one-period safe bond pays 1 next period and, by definition of the short rate i_t , costs e^{-i_t} today, so $\mathbb{E}_t[M_{t+1}] = e^{-i_t}$. On the other hand, m_{t+1} is normal with mean μ_t and variance $\lambda_t' \lambda_t$, since ε_{t+1} has mean zero and identity covariance. For any normal Z , $\mathbb{E}[e^Z] = e^{\mathbb{E}[Z] + \frac{1}{2} \text{Var}(Z)}$, so $\mathbb{E}_t[M_{t+1}] = e^{\mu_t + \frac{1}{2} \lambda_t' \lambda_t}$; equating the two expressions and taking logs gives μ_t .

³Cieslak and Schrimpf (2019) additionally include expected inflation in F , with inflation risk compensated. The qualitative implications I derive from my model are similar to those obtained in their paper.

is no consumption in the model, but g is the state that carries such news, and it is a simple AR(1), so neither ε^i nor ε^x moves it.

The second restriction is what makes risk premia time-varying. Growth risk earns $\lambda_{gx}x_t$ per unit of exposure, which is zero on average since x_t is mean-zero;⁴ were the price of risk constant, premia would be constant too and the fourth shock would not exist. The high-frequency literature often treats risk premia as fixed within the announcement window, and the model here does not. Empirical proxies for risk appetite (Bekaert et al., 2022; Bauer et al., 2023) are typically countercyclical; here x is defined as the component of risk appetite independent of g , so its cyclical part is absorbed into the growth state.

2.3 Information, timing, notation

Periods $t, t+1, \dots$ index FOMC announcements, one per period. For simplicity, no shocks occur between two consecutive announcements: all period- t innovations are realized and revealed at the period- t announcement.⁵ Information immediately before announcement t is $\mathcal{I}_{t-1}$ and contains all states dated $t-1$ and earlier. For any variable W , $\mathbb{E}_t^M[W_{t+1}]$ denotes the market's forecast just before the announcement $t+1$, where $\mathbb{E}^M$ is expectation under the market's beliefs. Denote $\Delta W_t \equiv W_t - \mathbb{E}_{t-1}^M[W_t]$ the announcement revision.

Key assumptions. At each announcement the market observes the full state $F_t = (g_t, i_t, x_t)$ ⁶ and knows the model structure and every parameter except the growth-response coefficient a , on which it holds the fixed belief $a^M < a$ (Bauer and Swanson, 2023a). Shocks are not observed directly but inferred from the state. The processes for g and x involve no unknown coefficient, so $\varepsilon_t^g = (g_t - \phi_g g_{t-1})/\sigma_g$ and $\varepsilon_t^x = (x_t - \phi_x x_{t-1})/\sigma_x$ are recovered exactly. The market observes g_t and $i_t - \phi_i i_{t-1} = a g_t + \sigma_i \varepsilon_t^i$, but not a : attributing only $a^M g_t$ to the growth response, it reads the policy shock as $\varepsilon_t^i + (a - a^M) g_t / \sigma_i$ rather than ε_t^i .

Fixed misperception. The market forms expectations of the policy rate under a misspecified model in which the growth-response coefficient is a known constant $a^M < a$, so the gap in coefficients is:

$$b \equiv a - a^M > 0.$$

Equivalently, its prior on a is fixed at a^M and never revised at the announcement: b is a primitive of the market's model, not a quantity it estimates. Thus, under $\mathbb{E}$ the coefficient

⁴Because I identify shocks from within-window announcement surprises, the level of the unconditional premium does not matter, and x_t can in principle have non-zero unconditional mean.

⁵News can in principle arrive between meetings; in this case it would be incorporated into the $t-1$ information set. For simplicity I assume this away.

⁶Equivalently, it observes the yield curve. Section 2.4 shows yields are affine in F_t with loadings the market knows, so three maturities invert to the three states; no privileged information about g or x is needed.

is the true a , which generates the data, and under $\mathbb{E}^M$ it is a^M , which generates prices and forecasts. In other words, I take a^M as a primitive of the pricing problem: the model takes the belief as given and does not describe how it is formed or maintained. The persistent *ex post* predictability of policy surprises documented by Bauer and Swanson (2023a) and evidence of a market that systematically underestimates the Fed’s growth response support this assumption.⁷

2.4 Asset-price responses and identification

2.4.1 Bond yields

Under the model assumptions, outlined above, zero-coupon n -period bond yield ($y_t^{(n)}$) is affine in the state,

$$y_t^{(n)} = A_n + B_n' F_t, \quad B_n = (B_n^g, B_n^i, B_n^x)' \in \mathbb{R}^3,$$

with A_n a scalar.

Each yield splits into the average of expected future short rates and a term premium, so its window response ($\Delta y_t^{(n)} = y_t^{(n)} - \mathbb{E}_{t-1}^M[y_t^{(n)}]$) splits the same way,

$$\Delta y_t^{(n)} = \underbrace{\frac{1}{n} \sum_{j=0}^{n-1} \Delta \mathbb{E}_t^M[i_{t+j}]}_{\text{expected future short rates}} + \underbrace{\Delta TP_t^{(n)}}_{\text{term premium}}.$$

Both pieces are governed by the loadings B_n (Appendix 9.6.1):

$$\frac{1}{n} \sum_{j=0}^{n-1} \Delta \mathbb{E}_t^M[i_{t+j}] = B_n^i \Delta i_t + B_n^g \sigma_g \varepsilon_t^g, \quad \Delta TP_t^{(n)} = B_n^x \sigma_x \varepsilon_t^x.$$

Expected rates move with the the policy rate surprise (Δi_t) and with growth news (ε_t^g); the term premium moves the revision in the price-of-risk factor (ε_t^x) and nothing else. Setting the price of risk to zero ($\lambda_{gx} = 0$) gives $B_n^x = 0$ and removes the premium entirely.

The announcement moves the state F and not the loadings, so differencing $y_t^{(n)} = A_n +$

⁷A rational agent would learn a from the data, and Bauer et al. (2024) document that perceived policy rules do adjust in response to policy surprises. Fixing a^M is therefore a simplification and a departure from full-information rational expectations (FIRE) assumption. What the model needs is only that a^M not change at the announcement itself; variation between announcements would not change any of the model’s implications.

$B'_n F_t$ gives

$$\Delta y_t^{(n)} = B'_n \Delta F, \quad \Delta F \equiv F_t - \mathbb{E}_{t-1}^M[F_t] = (\sigma_g \varepsilon_t^g, \Delta i_t, \sigma_x \varepsilon_t^x)'$$

Every maturity responds through the same three state revisions, and all three elements of ΔF follow directly from the dynamics of Section 2.1. Growth and the risk factor are AR(1), so their revisions are the shocks themselves; the rate revision follows from the policy rule and the market's forecast of it,

$$i_t = \phi_i i_{t-1} + a[\phi_g g_{t-1} + \sigma_g \varepsilon_t^g] + \sigma_i \varepsilon_t^i, \quad \mathbb{E}_{t-1}^M[i_t] = \phi_i i_{t-1} + a^M \phi_g g_{t-1},$$

whose difference is:

$$\Delta i_t = \underbrace{b \phi_g g_{t-1}}_{\text{FRN}} + \underbrace{a \sigma_g \varepsilon_t^g}_{\text{CBI}} + \underbrace{\sigma_i \varepsilon_t^i}_{\text{MP}}.$$

Substituting the three revisions,

$$\Delta y_t^{(n)} = \underbrace{B_n^i b \phi_g g_{t-1}}_{\text{FRN}} + \underbrace{\sigma_g (B_n^g + a B_n^i) \varepsilon_t^g}_{\text{CBI}} + \underbrace{\sigma_i B_n^i \varepsilon_t^i}_{\text{MP}} + \underbrace{\sigma_x B_n^x \varepsilon_t^x}_{\text{RP}}.$$

The first three terms are the expected-rates component and the fourth is the premium: MP, CBI and FRN affect yields only by moving the expected path of the short rate, while RP moves only the premium. The shortest maturity is the special case: $y_t^{(1)} = i_t$ and $B_1^i = 1$, $B_1^g = B_1^x = 0$, so $\Delta y_t^{(1)}$ is exactly the rate revision displayed above, the model's counterpart of equation (1). RP does not move the policy rate and enters only at $n \geq 2$ through B_n^x . This single zero is the reason the fourth shock can be invisible to a system built on the policy rate surprise and dominant in the ten-year yield, which is what Section 5 finds in the data.

The four shocks are mutually orthogonal: $\varepsilon^g, \varepsilon^i, \varepsilon^x$ are orthogonal by construction, and the FRN component $b \phi_g g_{t-1}$ is known before the announcement, hence orthogonal to the period- t innovations. Being a function of g_{t-1} , FRN is predictable from information available before the announcement, while MP and CBI, as period- t innovations, are not. Following Bauer and Swanson (2023a), the monetary policy surprise MPS is decomposed into a predictable component and a residual,

$$\text{MPS}_t = \underbrace{\hat{\beta}' y_t}_{\text{FIT}_t} + \underbrace{\hat{\varepsilon}_t}_{\text{RES}_t}, \quad (2)$$

where MPS_t is chosen to be 2-year bond yield surprise⁸, y_t is a vector of public information known prior to the announcement. This decomposition places FRN in FIT and other shocks in RES. The two enter the identification of Section 3 as separate observables.

Which shock moves which yield, and by how much, is therefore governed entirely by the loadings B_n . Define $S_n(\phi) \equiv 1 + \phi + \dots + \phi^{n-1} = (1 - \phi^n)/(1 - \phi)$; the closed forms are

$$B_n^i = \frac{S_n(\phi_i)}{n}, \quad B_n^g = \frac{a^M \phi_g}{n} \frac{S_n(\phi_g) - S_n(\phi_i)}{\phi_g - \phi_i}, \quad B_n^x = -\frac{\lambda_{gx} \sigma_g}{n} \sum_{k=1}^{n-1} \phi_x^{n-1-k} D_k,$$

where $D_k > 0$ is increasing in k ; its closed form, with the full derivation of the loadings, is in Appendix 9.6. The three loadings depend on different blocks of parameters: B_n^i on ϕ_i alone, B_n^g on (a^M, ϕ_g, ϕ_i) , and B_n^x on $(\lambda_{gx} \sigma_g, \phi_x)$ as well. The shapes the closed forms imply are:

$$\begin{aligned} B_n^i &> 0, \quad \text{strictly decreasing in } n, \\ B_n^g &\geq 0, \quad B_1^g = 0, \quad \text{hump-shaped,} \\ B_n^x &\leq 0, \quad B_1^x = 0, \quad |B_n^x| \text{ rising to an interior peak, then decaying to 0.} \end{aligned}$$

Under the normalization that a positive realization means $\varepsilon_t^i > 0$ a tightening, $\varepsilon_t^g > 0$ good growth news, $\varepsilon_t^x > 0$ a rise in the price of risk, and, since $b, \phi_g > 0$, g_{t-1} above its mean for FRN, the sign of each yield response is the sign of the loading it carries: B_n^i for MP and FRN, the composite $B_n^g + a B_n^i$ for CBI, and B_n^x for RP. The first two are strictly positive at every maturity, so MP, CBI and FRN raise yields throughout; $B_n^x \leq 0$, so RP lowers them.

2.4.2 Stocks

The log price–dividend ratio is affine in the same state, $pd_t = A_s + B_s' F_t$, with A_s a scalar and $B_s = (B_s^g, B_s^i, B_s^x)'$. Within the window dividends are predetermined, and the stock price response around the announcement can be written as:

$$\Delta \log P_t = \underbrace{B_s^g \sigma_g \varepsilon_t^g + B_s^i \Delta i_t}_{\text{expected cash flows and short rates}} - \underbrace{\sum_{j \geq 0} \kappa_1^j \Delta \mathbb{E}_t^M [EP_{t+j}]}_{\text{equity premium}},$$

where EP_{t+j} is the one-period equity premium, and κ_1 is Campbell–Shiller’s log-linearization constant, slightly lower than 1. As for the yield curve the premium term depends on the risk factor only: it equals $B_s^x \sigma_x \varepsilon_t^x$, and $\lambda_{gx} = 0$ removes it.

⁸In original paper, the first principal component of the first four euro-dollar futures is used for MPS. Here I depart from this slightly to be able to directly compare my identification with the Cieslak and Schrimpf (2019), which also use 2-year rate as one of their observables.

The announcement shocks moves the state variables, so $\Delta \log P_t = B'_s \Delta F$. Substituting,

$$\Delta \log P_t = \underbrace{B_s^i b \phi_g g_{t-1}}_{\text{FRN}} + \underbrace{\sigma_g (B_s^g + a B_s^i) \varepsilon_t^g}_{\text{CBI}} + \underbrace{\sigma_i B_s^i \varepsilon_t^i}_{\text{MP}} + \underbrace{\sigma_x B_s^x \varepsilon_t^x}_{\text{RP}},$$

we get similar decomposition with B_s in place of B_n : MP, CBI and FRN affect stocks through expected cash flows and rates, RP only through the equaity premium.

Which shock moves stocks, and by how much, is therefore governed by the vector B_s ; Appendix 9.7 solves for the closed form expressions of elements in B_s . Only $B_s^i < 0$ holds for all parameter values, so MP and FRN lower stock prices within announcement window. The other two signs depend on the parameters: CBI raises stock prices ($B_s^g + a B_s^i > 0$) and RP lowers them ($B_s^x < 0$) provided the policy rate rises less than one for one with real activity in the long run. Writing $\gamma \equiv a/(1 - \phi_i)$ for that policy response to growth, $\gamma^M \equiv a^M/(1 - \phi_i)$ for its perceived counterpart, and using $\kappa_1 \approx 1$ at announcement frequency, the conditions are

$$\gamma^M < 1 \quad \text{for } RP, \quad (1 - \phi_g) \gamma + \phi_g \gamma^M < 1 \quad \text{for } CBI.$$

Empirical estimates put the long-run response of the policy rate to real activity at or just below one (Carvalho et al., 2021). The market perceives less than this, $\gamma^M < \gamma$, so both conditions hold, the second binding. Equity is exposed to growth risk while bonds hedge it, so RP lowers stock prices while yields fall.

2.5 Summary of model results

Table 1 collects the results of the model and shows how the four shocks move the observables. Columns are the shocks and rows are the observables: RES and FIT, the unpredictable and predictable components of the policy surprise from equation (4.2); the S&P500 surprise; and the yield-curve slope, $10Y - 2Y$.

Table 1: Model-implied signs of the window responses

observable	MP (ε^i)	FRN ($b\phi_g g_{t-1}$)	CBI (ε^g)	RP (ε^x)
RES	+	0	+	(-)
FIT	0	+	0	0
$\Delta S\&P500$	-	-	+*	-**
ΔR_{10Y}	+	+	+	-
$ \Delta R_{10Y} $ vs $ \Delta R_{2Y} $	$<^\dagger$	$<^\dagger$	$<^\dagger$	$>^\dagger$

Notes. Normalization: $\varepsilon^i > 0$, $b\phi_g g_{t-1} > 0$, $\varepsilon^g > 0$, $\varepsilon^x > 0$. Most entries hold for any parameter values; some require additional conditions. *CBI requires $B_s^g + aB_s^i > 0$ and **RP requires only $B_s^x < 0$; the first implies the second, and both hold for typical Taylor rule estimates. † The two slope entries imply conditions on persistence, and for a typical calibration these signs hold.

Figure 1 plots the model’s announcement window yield response to each shock against maturity, at the following calibration: eight announcements per year, $\phi_g = 0.965$, $\phi_x = 0.889$, $\phi_i = 0.889$, $\gamma = a/(1 - \phi_i) = 0.79$ and $\gamma^M = 0.9\gamma$. The growth and price-of-risk persistences come from the calibration of Cieslak and Povala (2015); γ lies in the range of estimated Taylor coefficients (Carvalho et al., 2021), though any value below 1 satisfies the restrictions. Since $\lambda_{gx}\sigma_g$ scales B_n^x linearly and enters no condition, it is set to one.

All four shocks flatten the yield curve, but through affecting different parts of the curve: MP, FRN raise yields by more at the short end, CBI raises yields more at the middle, while RP lowers them by more at the long end. The first three shocks act only on expected future short rates, and since $\phi_i < 1$, shorter rates move more than long ones. RP shock moves the term premium, which accumulates with maturity, so longer yields fall by more than short ones. Figure 18 in Appendix 9.6 traces the range of ϕ_i over which the CBI and RP (†) entries hold jointly. Section 3.3 imposes these restrictions in the empirical model.

Two features of the table do the identifying work. First, MP and FRN are proportional in every asset price and differ only in predictability, so no sign restriction separates them. They are separated instead by which part of the surprise they move more, RES or FIT, which makes the Bauer–Swanson split a necessary feature of the identification. FRN depends on g_{t-1} and is therefore serially correlated, but no further condition on that serial correlation is needed: FIT is a projection on persistent pre-announcement variables, so whatever persistence FRN carries is inherited from that projection rather than imposed on top of it. Second, RP is the only shock with a *yields-down*, *stocks-down* signature, and the maturity dimension is the only observable that isolates it from MP shock.

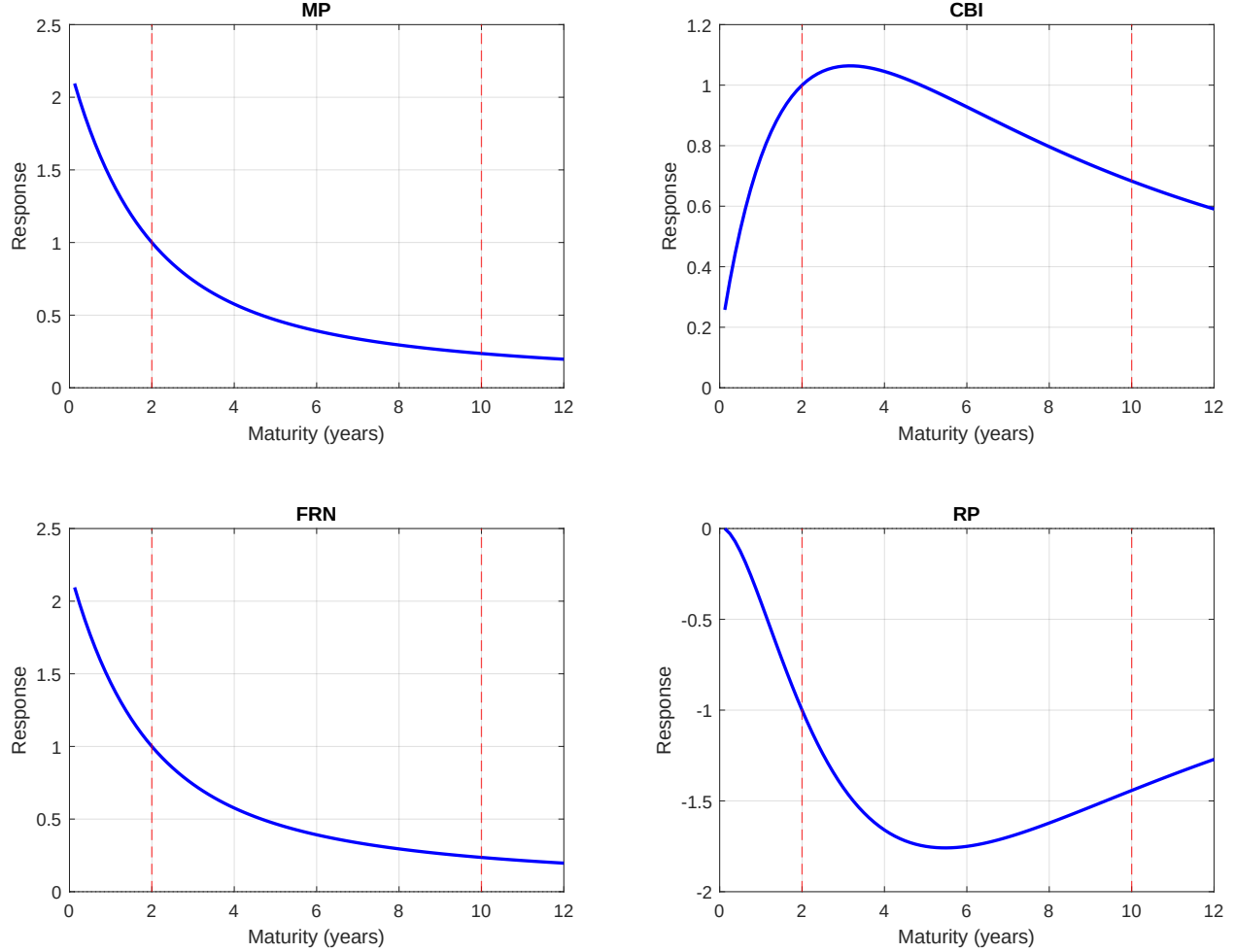


Figure 1: Model-implied window responses across the maturity curve

Notes. Each panel plots the model's window yield response to one shock against maturity, at the calibration of Subsection 2.5. Responses are scaled by the magnitude of the two-year response, so every curve passes through +1 (or -1 for RP) at the two-year point. Dashed lines mark the two- and ten-year maturities from which the slope is formed. Each slope entry of Table 1 holds when the curve lies below its two-year value at ten years.

2.6 Remarks

Three remarks close the model section. First, the model is small by construction and has obvious limitations. For instance, growth factor does not respond to policy, the market's belief about the policy rule never updates, and growth is the only priced risk. What I take from the model is only *qualitative* predictions: the signs in Table 1, and the effects of the shocks across maturities, hold for any parameter values or under relatively mild conditions. What it does not deliver is magnitudes of the shocks. The sizes of the four shocks are free parameters, so the model cannot say how the variance of an announcement divides among

them. Thus, how much each shock matters in the announcement window is the empirical question.

Second, the variation of an omitted shock goes to whichever remaining shock moves prices the same way. The FRN shock moves prices exactly as the monetary policy shock does and differs only in predictability, so a system without FRN sends its variation into the policy shock and leaves the other shocks unchanged. Jarociński and Karadi (2026) find that without the FRN shock, the monetary policy shock implies a larger and more persistent response of the one-year yield. That the variation goes to the policy shock rests on one feature of the model: the predictable part of the rate surprise carries no news about the economy (no ε_t^g), so *within the announcement window* stock prices fall and FRN is indistinguishable from a policy shock. It is, however, different, when it comes to monthly data. FRN component is high when growth has been high, and growth is persistent, so good growth yesterday means good growth ahead. Thus, a researcher who sees a rate hike followed by a boom concludes that the Fed knew something positive, and calls it the central bank information shock. This is the argument of Bauer and Swanson (2023a), and Jarociński and Karadi (2026) confirm it: their FRN component predicts a temporary boom, much like the CBI shock. I later show this is also true for the empirical identification with all four shock included at the same time.

Third, a risk premium shock lowers stock prices and yields together, the mirror image of good news about the economy, so an omitted risk premium shock is attributed to the information shock with the sign reversed. The resemblance is not exact, in contrast to the case of FRN: an information shock moves medium yields more than long ones, while the risk premium shock works through the term premium, which accumulates with maturity; this difference is the reason the ten-year yield is part of the observable set. The absorption of RP by CBI is therefore partial, and its degree depends on the variance of the risk premium shock relative to the other shocks, which the model does not restrict. The implications of RP shock omission are therefore an empirical question.

3 Identification

Having set out the model predictions, I now turn to the empirical identification. High-frequency surprises around FOMC announcements reflect a mixture of structural shocks. Let u_t denote the vector of observed reduced-form high-frequency surprises and ε_t the vector of structural shocks, linked by

$$u_t = B \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I, \quad (3)$$

so that $BB' = \Sigma_u$. Since B is not uniquely pinned down by Σ_u , identification requires additional restrictions. The two schemes below, Jarociński and Karadi (2026) and Cieslak and Schrimpf (2019), achieve this by choosing different observables and different restrictions. I then introduce JKCS, which combines the two, and is based entirely on the model’s restrictions, and state the choices that make the three sets of shocks comparable. Section 4 describes the surprises themselves and every other series the paper uses.

3.1 Jarocinski–Karadi (2026)

JK identify three shocks, monetary policy (MP), Fed response to news (FRN), and central bank information (CBI), from three high-frequency observables.

The reduced-form surprise vector is $u_t = (\text{RES}_t, \text{FIT}_t, \Delta\text{S\&P500}_t)'$, and B takes the form

$$\begin{pmatrix} \text{RES} \\ \text{FIT} \\ \Delta\text{S\&P500} \end{pmatrix} = \underbrace{\begin{pmatrix} + & a_{12} & a_{13} \\ a_{21} & + & a_{23} \\ - & - & + \end{pmatrix}}_{B^{\text{JK}}} \begin{pmatrix} \varepsilon^{\text{MP}} \\ \varepsilon^{\text{FRN}} \\ \varepsilon^{\text{CBI}} \end{pmatrix}, \quad (4)$$

subject to

$$a_{13} + a_{23} > 0, \quad |a_{21}| < a_{11}, \quad |a_{12}| < a_{22}. \quad (5)$$

The sign restrictions encode the following co-movements. An MP shock generates negative co-movement between RES and $\Delta\text{S\&P500}$; an FRN shock generates negative co-movement between FIT and $\Delta\text{S\&P500}$; a CBI shock generates positive co-movement between $\text{MPS} = \text{RES} + \text{FIT}$ and $\Delta\text{S\&P500}$, which delivers the joint restriction $a_{13} + a_{23} > 0$. The magnitude restrictions formalize the “good proxy” idea: the MP shock loads more on RES than on FIT ($|a_{21}| < a_{11}$), and the FRN shock loads more on FIT than on RES ($|a_{12}| < a_{22}$).⁹ JK sharpen the set-identified posterior by exploiting heteroskedasticity between FOMC and non-FOMC events, but the identifying content is already contained in (4)–(5). I implement only the sign- and magnitude-restricted block and leave the heteroskedasticity step to future work.

3.2 Cieslak–Schrimpf (2019)

CS identify three shocks, growth (G), monetary policy (MP), and risk premium (RP), from the high-frequency reactions of the two- and ten-year Treasury yields and stock returns,

⁹Since FIT_t and RES_t are orthogonal by (4.2), $\sum_{j=1}^3 a_{1j}a_{2j} = 0$ holds automatically in the system, keeping them uncorrelated.

$u_t = (\Delta R_{2Y,t}, \Delta R_{10Y,t}, \Delta S\&P500_t)'$. The matrix B satisfies

$$\begin{pmatrix} \Delta R_{2Y} \\ \Delta R_{10Y} \\ \Delta S\&P500 \end{pmatrix} = \underbrace{\begin{pmatrix} + & + & - \\ + & + & - \\ + & - & - \end{pmatrix}}_{B^{CS}} \begin{pmatrix} \varepsilon^G \\ \varepsilon^{MP} \\ \varepsilon^{RP} \end{pmatrix}. \quad (6)$$

The sign restrictions reflect standard asset-pricing reasoning. A positive growth shock raises expected cash flows and pushes the real rate up, so stocks and yields move together;¹⁰ a monetary tightening raises yields and lowers stock prices, so they move in opposite directions; and a positive risk premium shock lowers both stock prices and yields, as government bonds become valuable in flight-to-safety episodes.

The RP shock is a change in the compensation investors require for bearing risk. Such shocks are not specific to FOMC announcements. They are, however, expected within the narrow FOMC window because a FOMC announcement can affect policy uncertainty and carries information about risks to the outlook. The shock can capture shifts in sentiment or effective risk aversion that move equity prices and bond yields in the same direction, independently of news about the economy, or the policy rate (Cieslak and Schrimpf, 2019). A positive risk premium shock, a risk-off event, lowers stocks and flattens the yield curve, and its effect loads predominantly on the long end of the term structure through the bond risk-premium channel.

The sign restrictions in (6) alone cannot separate growth from risk premium shocks, since both generate positive stock–yield co-movement. To distinguish them, CS exploit the maturity dimension of the yield curve through monotonicity restrictions:

$$|b_{2Y,RP}| < |b_{10Y,RP}|, \quad (7)$$

$$|b_{2Y,G}| > |b_{10Y,G}|, \quad (8)$$

$$|b_{2Y,MP}| > |b_{2Y,RP}|, \quad (9)$$

$$|b_{10Y,MP}| < |b_{10Y,RP}|, \quad (10)$$

where $b_{n,s}$ denotes the loading of the n -maturity yield on shock s . Growth shocks move the two-year yield more than the ten-year, since their effect runs through expected short rates and decays with maturity, while risk premium shocks move the ten-year more, through the term premium. CS note that (7) and (8) are essential for separating the two shocks.

¹⁰The growth shock of Cieslak and Schrimpf (2019) is therefore the counterpart of the CBI shock of Jarociński and Karadi (2026), and I refer to it as CBI throughout.

3.3 JKCS: joint identification

Neither JK nor CS identifies all four shocks. JK omit the risk premium shock, so risk-premium variation is attributed to MP, FRN or CBI. CS omit the response-to-news shock and attribute its variation to the remaining three.

JKCS combines the two systems. It takes the RES/FIT/ Δ S&P500 block from JK and adds the ten-year Treasury yield as a fourth observable. A single long-maturity yield, rather than the two- and ten-year yields separately, keeps the system square, four observables and four shocks, while retaining the maturity dimension that CS show is necessary to separate RP from the other shocks. The two-year surprise is not discarded: under the measurement convention of Section 4 it is the sum RES + FIT. The identifying equation is

$$\begin{pmatrix} \text{RES} \\ \text{FIT} \\ \Delta\text{S\&P500} \\ \Delta R_{10Y} \end{pmatrix} = \underbrace{\begin{pmatrix} + & c_{12} & + & - \\ c_{21} & + & c_{23} & c_{24} \\ - & - & + & - \\ + & + & + & - \end{pmatrix}}_{B^{\text{JKCS}}} \begin{pmatrix} \varepsilon^{\text{MP}} \\ \varepsilon^{\text{FRN}} \\ \varepsilon^{\text{CBI}} \\ \varepsilon^{\text{RP}} \end{pmatrix}, \quad (11)$$

where c_{mj} is the response of observable m to shock j . The restrictions are of three kinds: signs, magnitude conditions, and one external correlation restriction. All but the last are direct implications of the model of Section 2.

The entries of (11) represent the first layer of restrictions. The policy and response-to-news shocks raise yields and lower stock prices; the information shock raises both, which is what distinguishes it; and the risk premium shock lowers the unpredictable component of the surprise, stock prices and the ten-year yield together. The blank entries are the responses the model leaves free. Apart from sign restrictions in (11), the two-year rate (RES + MPS) carries four additional sign restrictions, indicating how the four shocks affect the 2-year yield surprise, identical to the restrictions in the fourth row:

$$c_{11} + c_{21} > 0, \quad c_{12} + c_{22} > 0, \quad c_{13} + c_{23} > 0, \quad c_{14} + c_{24} < 0. \quad (12)$$

Next, additional magnitude restrictions separate shocks that the signs cannot. Each shock moves one component of the policy surprise more than the other, and RES and FIT each respond more to one shock than to another,

$$|c_{21}| < c_{11}, \quad |c_{12}| < c_{22}, \quad |c_{23}| < c_{13}, \quad |c_{24}| < |c_{14}|, \quad |c_{12}| < c_{11}, \quad |c_{21}| < c_{22}. \quad (13)$$

The first four say that MP, CBI and RP move RES more than FIT while FRN moves FIT

more: FIT is the component the response-to-news shock dominates. The last two tells that MP shock affects RES more than FIT, and vice versa for FRN shock.

Now, even with all these restrictions above imposed some shocks would be indistinguishable from each other. That is why I use additional magnitude restrictions which restrict the effects of the shocks the over the two maturities. The MP, FRN and CBI shocks move the two-year rate by more than the ten-year yield, while the risk premium shock moves the ten-year by more, resulting in:

$$|c_{1j} + c_{2j}| > |c_{4j}| \quad (j = MP, FRN, CBI), \quad |c_{14} + c_{24}| < |c_{44}|, \quad (14)$$

$$|c_{11} + c_{21}| > |c_{14} + c_{24}|, \quad |c_{41}| < |c_{44}|. \quad (15)$$

Finally, to make sure that RP shock is separated from CBI shock (the only thing that separates them so far is the magnitude restrictions above), the risk premium shock is required to correlate negatively with the change in the Bauer et al. (2023) risk-appetite index,¹¹

$$\text{corr}(\varepsilon_t^{RP}, \Delta BBM_t) < 0. \quad (16)$$

This is the only external restriction in the model, which uses information from outside the announcement window. Table 2 summarizes the full set of identifying restrictions in all three systems.

¹¹The change is constructed as $\Delta BBM_t = BBM_{t+1} - BBM_{t-1}$, a two-day window centered on the announcement day. A positive ΔBBM_t is a risk-on move around the announcement, and the risk premium shock is normalized as a risk-off shock. This index is chosen because it is available at daily frequency over the full FOMC sample and is constructed from a wide range of risk indicators drawn from equity, fixed-income, credit, and FX markets.

Table 2: Identifying restrictions: the JK, CS, and JKCS systems

	ε_t^{MP}	ε_t^{FRN}	ε_t^{CBI}	ε_t^{RP}	Magnitude and other restrictions	
Panel A: JK system						
RES	+	free	free	n.a.	$b_{2Y,CBI} > 0$	
FIT	free	+	free	n.a.	$ b_{\text{FIT},MP} < b_{\text{RES},MP}$	
$\Delta\text{S\&P500}$	−	−	+	n.a.	$ b_{\text{RES},FRN} < b_{\text{FIT},FRN}$	
Panel B: CS system						
$\Delta\text{S\&P500}$	−	n.a.	+	−	$ b_{2Y,CBI} > b_{10Y,CBI} $	$ b_{2Y,RP} < b_{10Y,RP} $
ΔR_{2Y}	+	n.a.	+	−	$ b_{2Y,MP} > b_{2Y,RP} $	$ b_{10Y,MP} < b_{10Y,RP} $
ΔR_{10Y}	+	n.a.	+	−		
Panel C: JKCS system						
RES	+	free	+ [§]	− [§]	$ b_{\text{FIT},MP} < b_{\text{RES},MP}$	$ b_{\text{RES},FRN} < b_{\text{FIT},FRN}$
FIT	free	+	free	free	$ b_{\text{RES},FRN} < b_{\text{RES},MP}^{\S}$	$ b_{\text{FIT},MP} < b_{\text{FIT},FRN}^{\S}$
$\Delta R_{2Y} = \text{RES} + \text{FIT}$	+	+ [§]	+	−	$ b_{\text{FIT},CBI} < b_{\text{RES},CBI}^{\S}$	$ b_{\text{FIT},RP} < b_{\text{RES},RP} ^{\S}$
$\Delta\text{S\&P500}$	−	−	+	−	$ b_{2Y,MP} > b_{10Y,MP} ^{\S}$	$ b_{2Y,FRN} > b_{10Y,FRN} ^{\S}$
ΔR_{10Y}	+	+ [§]	+	−	$ b_{2Y,CBI} > b_{10Y,CBI} $	$ b_{2Y,RP} < b_{10Y,RP} $
					$ b_{2Y,MP} > b_{2Y,RP} $	$ b_{10Y,MP} < b_{10Y,RP} $
					$\text{corr}(\varepsilon_t^{RP}, \Delta\text{BBM}_t) < 0^{\S}$	

Notes: $b_{m,j}$ is the impact response of observable m to shock j . In Panel C, $b_{2Y,j} \equiv b_{RES,j} + b_{FIT,j}$, so the two-year conditions are internal inequalities on the impact matrix rather than restrictions on a fifth observable. “n.a.” means the shock is not identified in that system; “free” means the entry carries no sign restriction. In each system one sign per column is a normalization that labels the shock. JK also sharpen inference using heteroskedasticity between FOMC and non-FOMC events, which is not implemented here.

[§] Imposed here but in neither 3-by-3 system.

Estimation of all three systems follows the rejection method of Rubio-Ramirez et al. (2010): a candidate impact matrix $B = PQ'$, with P the lower Cholesky factor of the reduced-form covariance matrix and Q orthogonal, is kept when it satisfies every restriction of that system. The accepted draws are the identified set, and point estimates are at the Fry–Pagan rotation, the draw closest to the element-wise median of B (Fry and Pagan, 2011). Appendix 9.1 states the algorithm.

3.4 Making the three systems comparable

The paper compares three sets of shocks, so they must be estimated on identical samples of policy events, with any difference between them attributable to the identification assumptions rather than to the inputs. Below are the choices that ensure this comparability.

1. Sample. All three systems are estimated on the event sample of Section 4, which uses scheduled and unscheduled announcements together with the press conferences.

2. JK modifications. Jarociński and Karadi (2026) build the policy surprise from the first principal component of the four nearest eurodollar futures. To ensure comparability with the other two identification schemes, I use the two-year Treasury surprise instead, so that $\text{RES} + \text{FIT} = \Delta R_{2Y}$ and RES, FIT and $\Delta \text{S\&P500}$ are numerically the same series in all three systems. The cost of the choice is small. Under the two measurement conventions for MPS the JK shocks have correlations of 0.97 or above for MP and CBI shocks over the chosen sample. Because the sample excludes speeches and testimonies, I also do not use the heteroskedasticity-based part of the identification. Without it, all identification schemes are based on similar sign and magnitude restrictions, and neither borrows anything from additional assumptions or data. The remaining restrictions are those of Jarociński and Karadi (2026). The predictor set in (4.2) is the seventeen-variable set of Table 4.

3. CS as published. The Cieslak–Schrimpf system enters exactly as in the published paper, with all restrictions imposed there.

4 Data

The paper uses several types of data. First, high-frequency surprises around FOMC communication events, and public information released before each announcement is used for the identification of shocks. Second, to analyze the effects of the shocks outside the announcement windows, daily financial series and monthly macroeconomic indicators are used as the outcome variables.

4.1 Announcement windows

FOMC surprises come from the U.S. Monetary Policy Event-Study Database of Acosta et al. (2025). A statement surprise is the change in an asset price over a 30-minute window, from 10 minutes before the release to 20 minutes after. A press-conference surprise is the change over a 70-minute window, from 10 minutes before the Chair begins to 60 minutes after. The sample starts from January 1991 and ends in March 2025 and uses scheduled and unscheduled statements as well as press conferences. Some of the robustness checks additionally include the minutes releases. Extending the event set is left to future work. Table 3 reports the composition of the event sample. The identification uses 389 events on 306 announcement dates.

Table 3: The event sample

	Count
Announcement dates	306
Scheduled meetings	272
Unscheduled meetings	34
With a press conference	83
Statement windows	306
Press-conference windows	83
Windows used in the identification	389

Notes. January 1991 to March 2025. A statement and the press conference that follows it are two windows on the same date, so the window count exceeds the date count by the number of press conferences. The same event sample enters all three identification systems.

Four observables enter the JKCS system: the Bauer–Swanson residual (RES) and fitted value (FIT), constructed from the 2-year zero-coupon bond surprise and the seventeen predictors listed in Table 4; the S&P500 surprise $\Delta S\&P500$ from E-mini futures; and the ten-year Treasury yield surprise ΔR_{10Y} . The JK and CS systems use same variables.

4.2 Predictors of the policy surprise

The predictor set y_t in

$$\text{MPS}_t = \underbrace{\hat{\beta}' y_t}_{\text{FIT}_t} + \underbrace{\hat{\varepsilon}_t}_{\text{RES}_t},$$

follows Jarociński and Karadi (2026): it is the union of the regressors of Bauer and Swanson (2023a), Bauer and Swanson (2023b) and Swanson (2024), seventeen in all, see Table 4 for definitions.

Table 4: Predictors of the policy surprise

Predictor	Definition	Source
<i>Panel A. Bauer and Swanson (2023a)</i>		
Payroll growth	Twelve-month log change in nonfarm payrolls, to the latest release	PAYEMS, FRED
Treasury skewness	Implied skewness of the ten-year Treasury yield from options on ten-year note futures	Bauer and Chernov (2024) series
Stock returns	Three-month change in the log S&P 500	Bloomberg
Yield-curve slope	Three-month change in the second principal component of one- to ten-year zero-coupon Treasury yields	Federal Reserve Board (Gürkaynak et al. (2007))
Commodity prices	Three-month change in the log Bloomberg Commodity Index	Bloomberg
<i>Panel B. Bauer and Swanson (2023b)</i>		
Payrolls surprise	Released value minus the median forecast for that release	MMS survey, Haver Analytics
Unemployment surprise	As above, for the unemployment rate	As above
Real GDP surprise	As above, for real GDP; advance, second and third releases stacked by release date	As above
Core CPI surprise	As above, for core CPI	As above
Core CPI forecast	Median forecast for the core CPI release of the announcement month	As above
Core CPI acceleration	$200 (\ln P_{m-2} - 2 \ln P_{m-8} + \ln P_{m-14})$, with P the core CPI index	CPILFESL, FRED
Business cycle index	Brave–Butters–Kelley cyclical component, month $m-1$	BBKMCY, FRED
<i>Panel C. Swanson (2024)</i>		
Two-year yield	Three-month change in the two-year zero-coupon Treasury yield	Federal Reserve Board (Gürkaynak et al. (2007))
Ten-year yield	Three-month change in the ten-year zero-coupon Treasury yield	As above
Baa spread	Three-month change in the Moody’s Baa yield less the ten-year Treasury yield	BAA10Y, FRED
Shadow rate	Three-month change in the Wu–Xia shadow federal funds rate	Wu and Xia (2016) series
Financial conditions	Four-week change in the Adjusted National Financial Conditions Index	ANFCI, Chicago Fed

Notes. Several regressors appear in more than one of the three papers.

4.3 Other data

The local projections in Section 6.1 use twelve daily series: the two- and ten-year breakeven inflation rates, the five-year investment-grade and high-yield CDX total-return indices, and the S&P 500, all from Bloomberg; the VIX and the ICE BofA BBB corporate spread; the ten-year yield with its decomposition into an expectations component and a term premium from Adrian et al. (2013), from the Federal Reserve Bank of New York; the risk appetite index of Bauer et al. (2023); and the risk aversion index of Bekaert et al. (2022), the last two from the authors’ data pages. With one exception these series play no role in the

identification. The exception is the risk appetite index, which enters through the external correlation restriction.

The VAR in Section 6.2 uses five monthly variables: the one-year Treasury yield, the log S&P500 and the log PCE price index, all from FRED-MD; the excess bond premium of Gilchrist and Zakrajšek (2012), from the Federal Reserve Board’s updated series; and log real monthly GDP, interpolated to monthly data (S&P Global). The quarterly series is from the Bureau of Economic Analysis and the five indicators are from FRED-MD.

5 What the four shocks explain within the window

Having set out the identifying restrictions and the data, I turn to what the four shocks explain: how much of each surprise they account for, and what it costs to leave one out, as the JK and CS systems do. All three systems are estimated on the event sample of Section 4, so no difference reported below comes from the sample.

Write $u_t = B\varepsilon_t$ for the vector of observed window surprises, and let b_{mj} be the response of surprise m to a one-standard-deviation shock j . The shocks are orthogonal with unit variance, so the variance of surprise m is $\sum_k b_{mk}^2$, and the share explained by shock j is

$$s_{mj} = \frac{b_{mj}^2}{\sum_k b_{mk}^2}.$$

Shares sum to one across shocks by construction. The system has as many observables as shocks, so nothing is left unexplained: all variation in the observables is attributed to the shocks. What the identifying restrictions decide instead is how the window variance is allocated across the four shocks.

5.1 Omitting the Fed-response-to-news shock

Table 5 shows the four-shock variance decomposition of observable surprises next to the three-shock CS system, with the Δ column showing the difference between them. The last column reports how often the difference keeps the sign it has at the reported point, as the four-shock system varies over its identified set and CS is held at its own Fry–Pagan rotation.

First, all four shocks are present in the data, and none becomes negligible once they are identified together, but which one dominates depends on the observable. The policy shock is the largest contributor to the equity surprise, at 60.6 %. Two-year rate surprise is explained mainly by policy shock (39.5%) and CBI shock (36.1 %). The risk premium shock is the main driver of the ten-year surprise, at 64.2 %, more than the other three combined.

Adding the response-to-news shock changes the contributions everywhere, including at the long end of the yield curve. The FRN shock takes 8.0 % of equity variance, 21.2 % of the two-year rate and 10.7 % of the ten-year rate. The policy shock’s share falls in all three observables once the FRN shock is added, by 13.9, 11.8 and 7.4 points for the S&P500, the two-year rate and the ten-year rate. In each panel that is the largest decline, and it is the direction the model predicts. These three declines are also the most stable entries in the table: each keeps its sign in about three quarters of the identified set.

The other two shocks contributions are not changed in the same way. The information shock share rises 8.0 points for stock surprise, falls 7.2 for the two-year and rises 1.0 for the ten-year surprises, once the FRN is added. However, these changes are not robust, since they don’t hold for most of the draws in admissible set. The risk premium shares of explained variance are mainly not affected by adding FRN shock. Omitting the news shock therefore mainly distorts what is attributed to the policy shock, and not to the other shocks.

Table 5: Announcement-window variance decomposition: omitting the Fed-response-to-news shock

Shock	CS (3 shocks)	JKCS (4 shocks)	Δ (pp)	sign (%)
$\Delta S\mathcal{E}P500, \sigma = 0.83\%$				
MP	74.4 [24.4, 95.6]	60.6 [21.5, 90.1]	−13.9	77
FRN	n.a.	8.0 [0.1, 47.9]	+8.0	n.a.
CBI	19.6 [0.2, 73.8]	27.6 [0.5, 63.0]	+8.0	44
RP	6.0 [0.0, 28.4]	3.8 [0.0, 26.4]	−2.1	59
$\Delta R_{2Y}, \sigma = 5.33 \text{ bp}$				
MP	51.3 [15.3, 90.7]	39.5 [10.6, 70.5]	−11.8	75
FRN	n.a.	21.2 [1.6, 31.9]	+21.2	n.a.
CBI	43.4 [4.2, 81.7]	36.1 [8.7, 72.8]	−7.2	60
RP	5.3 [0.1, 24.7]	3.1 [0.1, 20.2]	−2.3	48
$\Delta R_{10Y}, \sigma = 4.55 \text{ bp}$				
MP	23.5 [5.2, 44.9]	16.1 [1.7, 35.6]	−7.4	76
FRN	n.a.	10.7 [0.3, 29.0]	+10.7	n.a.
CBI	7.9 [0.0, 31.8]	9.0 [0.1, 27.9]	+1.0	40
RP	68.6 [48.1, 90.0]	64.2 [47.1, 84.0]	−4.3	62

Notes. CS is the three-shock Cieslak–Schrimpf system as published; JKCS is the four-shock system mapped to the same three variables, where $RES + FIT = \Delta R_{2Y}$. Both decompose the same $\hat{\Sigma}$ over the same 389 windows. Point shares correspond to Fry–Pagan rotations; brackets show 5–95% ranges of each identified set.

5.2 Omitting the risk premium shock

Table 6 does the same decomposition for the Jarocinski–Karadi system, which has no risk premium shock. The risk premium shock accounts for less than 5 % of the variance of the residual, the fitted component, the equity surprise and the two-year rate. This does not mean that the RP shock is unimportant, because it mainly affects the ten-year rate surprise, which JK does not observe and which is therefore absent from the table. Adding the risk premium shock nonetheless changes how the other three divide the explained variance.

The largest single change is on the fitted component of MPS, where the CBI share falls by 20.0 points, of which the FRN shock takes 16.8. The model implies that FRN is the only shock in FIT, since it is a function of information known before the announcement while MP, CBI and RP are period- t innovations. The identification does not impose this, and the four-shock system attributes 87.9 % of FIT to FRN against 71.1 % in JK. The contribution of MP to the residual part and to the two-year rate surprise falls by 17.5 and 15.9 points. The policy shock share rises by 16.9 points for the S&P500 surprise, while the FRN share falls by 13.5. Adding the risk premium shock therefore changes the shares of all three remaining shocks, and by more than adding the FRN shock changes them in Table 5: the largest change among the shared shocks is 20.0 points here against 13.9 there, and the changes keep their sign more often across the rotations.

These differences cannot be read as the effect of the missing shock alone. JK is estimated with the restrictions used in its own paper, so it differs from the baseline in its restrictions as well as in the shocks it identifies. In addition to the restrictions in JK, the four-shock model requires the MP effect on *RES* to exceed the FRN effect, the FRN effect on *FIT* to exceed the MP effect, and the CBI shock to raise *RES* and to move it more than *FIT*. The same point, albeit to a lesser degree, applies to the comparison with CS: JKCS restricts the MP loadings by maturity and requires the risk premium shock to correlate negatively with the risk appetite index, and CS does neither. I re-estimate both three-shock systems under the restrictions used in the four-shock baseline in Appendix 9.4, and found that restrictions play little role in explaining the difference between variance shares in CS and JKCS models, while difference between JK and JKCS models are mainly due to difference in restrictions.

Table 6: Announcement-window variance decomposition: omitting the risk-premium shock

Shock	JK (3 shocks)	JKCS (4 shocks)	Δ (pp)	sign (%)
<i>RES</i> , $\sigma = 5.04$ bp				
MP	77.2 [15.8, 99.0]	59.7 [20.9, 88.4]	−17.5	82
FRN	0.0 [0.0, 7.6]	2.8 [0.0, 8.1]	+2.7	94
CBI	22.8 [0.1, 81.4]	34.9 [4.6, 72.5]	+12.1	66
RP	n.a.	2.7 [0.3, 22.1]	+2.7	n.a.
<i>FIT</i> , $\sigma = 1.73$ bp				
MP	7.2 [0.1, 39.7]	9.9 [0.3, 28.0]	+2.8	52
FRN	71.1 [11.3, 96.6]	87.9 [40.8, 96.2]	+16.8	69
CBI	21.7 [0.2, 80.3]	1.7 [0.0, 38.7]	−20.0	82
RP	n.a.	0.4 [0.0, 9.8]	+0.4	n.a.
$\Delta S\&P500$, $\sigma = 0.83\%$				
MP	43.7 [4.1, 92.2]	60.6 [21.5, 90.1]	+16.9	68
FRN	21.5 [0.2, 76.8]	8.0 [0.1, 47.9]	−13.5	72
CBI	34.8 [0.3, 81.2]	27.6 [0.5, 63.0]	−7.3	77
RP	n.a.	3.8 [0.0, 26.4]	+3.8	n.a.
ΔR_{2Y} , $\sigma = 5.33$ bp				
MP	55.4 [5.9, 96.0]	39.5 [10.6, 70.5]	−15.9	80
FRN	8.3 [0.1, 29.4]	21.2 [1.6, 31.9]	+12.9	71
CBI	36.3 [0.4, 85.3]	36.1 [8.7, 72.8]	−0.1	48
RP	n.a.	3.1 [0.1, 20.2]	+3.1	n.a.

Notes. JK is the three-shock Jarocinski–Karadi system with sign and magnitude restrictions as published. $\Delta R_{2Y} = RES + FIT$ is not a separate observable, and ΔR_{10Y} is not in table because JK model does not include it in observables. Both identification schemes decompose the same $\hat{\Sigma}$ over the same 389 windows. Point shares correspond to Fry–Pagan rotations; brackets show 5–95% ranges of each identified set.

5.3 What each three-shock system recovers

Next, I further show how the shocks of each three-shock system are related to the four shocks in the joint system. Both three-shock observable sets are nested in the JKCS set, so each shock of a three-shock system is an exact combination of the four JKCS shocks, $\tilde{\varepsilon}_j = \sum_i \lambda_{ij} \varepsilon_i$, with the weights satisfying $\sum_i \lambda_{ij}^2 = 1$. Table 7 reports $100\lambda_{ij}^2$, the share of each three-shock system’s shock accounted for by each JKCS shock. The four entries in any column sum to 100, and a column with a single entry close to 100 means the shock is the same in both systems, while other cases mean the shock is a combination of other shocks,

and the table says in what proportion.

Several results follow from the table. First, in every column the largest entry lies on the row of the same name, so the shock labels broadly agree across identification systems. That largest entry is above 90% in four of the six columns and near 79% in the other two. Second, a fifth of what the CS system labels a monetary policy shock is the FRN shock. Third, a fifth of the JK CBI shock is made up of the other three shocks. The largest part, 8.9%, is the RP shock, meaning that part of the measured information effect in an announcement window is, in fact, risk-premium variation. Finally, the last column, Σ , gives the row sum, which is the share of each JKCS shock that the three-shock system spans (i.e., the R^2 from regressing a JKCS shock on the three shocks of that system). The CS shocks explain only 80% of the policy shock, and JK shocks explain 90% of the information shock of the joint system.

Table 7: How shocks are related across identifications

<i>Panel A. JK system (3 shocks)</i>				
JKCS shock	JK MP	JK FRN	JK CBI	Σ
MP	96.2 [88.5]	0.2 [1.9]	3.5 [4.2]	99.9
FRN	0.0 [1.1]	91.6 [88.7]	8.3 [5.4]	99.9
CBI	3.7 [5.7]	7.5 [3.6]	79.3 [70.1]	90.5
RP	0.1 [1.0]	0.7 [1.5]	8.9 [13.0]	9.7
<i>Panel B. CS system (3 shocks)</i>				
JKCS shock	CS MP	CS CBI	CS RP	Σ
MP	79.4 [72.7]	0.0 [4.3]	0.1 [1.4]	79.5
FRN	19.9 [18.7]	3.6 [3.2]	0.5 [1.2]	24.0
CBI	0.7 [3.4]	95.5 [87.0]	0.6 [1.6]	96.8
RP	0.0 [1.0]	0.8 [1.7]	98.8 [93.5]	99.6

Notes. Entries are the share of each JK and CS shock accounted for by each JKCS shock, in %; the four entries in a column sum to 100. Σ is the row sum, the share of each JKCS shock the three-shock system explains. Points are Fry–Pagan rotations, brackets the median over the JKCS identified set.

Table 8 says the same thing in units of correlation. Correlations across the shocks from any two identification approaches are high, ranging from 0.86 to 0.99. The lowest correlations involve two shocks. The first is the CS MP shock against its counterparts in the other two systems, which the FRN shock contaminates; the second is the JK CBI shock against its counterparts, which the RP shock contaminates. For applied work the choice of system matters for these two of the four shocks and not for the other two.

Table 8: Pairwise correlations of shocks

Shock	JK–JKCS	JK–CS	JKCS–CS
MP	0.98	0.86	0.89
CBI	0.89	0.90	0.98
FRN	0.96	-	-
RP	-	-	0.99

Notes. Correlations on the same 389 windows, with each system at its Fry–Pagan rotation. “-” marks a shock absent from one of the two systems.

One property of the shocks is not imposed by any of the three systems, and can be used to check them. A structural shock should not be predictable from its own past. The Fed-response-to-news shock is the only exception, since it drives the component of the surprise that is predictable from public information. Hence, we can expect it to be autocorrelated. Table 9 reports the first-order autocorrelation across consecutive announcement days. The autocorrelation of the policy shock is not significantly different from zero in any of the three systems. The FRN shock is autocorrelated at the 1% significance level in both systems that identify it, which is what its definition implies. The CBI shock is where the systems differ: its autocorrelation is significant at the 5% level in JK, and not significant at the 10% level in either JKCS or CS. The JK omitting of RP shock therefore generates a CBI shock that is forecastable from its own past.

Table 9: First-order autocorrelation of the identified shocks

Shock	JK	JKCS	CS
MP	0.01	0.04	-0.05
CBI	0.15**	0.05	0.08
FRN	0.28***	0.37***	-
RP	-	-0.10*	-0.07

Notes. Autocorrelation is measured across consecutive FOMC announcements (306 days in total). Shocks are at the Fry–Pagan rotation of each system, with same-day windows summed, so the lag is one announcement day. Stars mark rejection of a Null of zero first-order autocorrelation at the 1, 5 and 10 % levels. “-”: a shock absent from that system.

5.4 The shocks and the Swanson (2021) factors

Next, I compare how the Swanson (2021) decomposition of each FOMC announcement news into a federal funds rate (FFR), a forward guidance (FG) and an asset-purchase (LSAP) factor is related to the identified shocks. The conjecture here is that the FFR factor is

mostly related to MP and FRN shocks, the FG factor is mainly related to CBI, while LSAP, due to its effects on the yield curve, should be related to the RP shock. Table 10 tests this empirically: each column regresses one of Swanson’s factors on the shocks, evaluated at the Fry–Pagan rotation, with same-day statement and press-conference windows summed to one observation per announcement date, over the 239 dates the two event sets share, which end in June 2019. The second and third column groups restrict the sample to scheduled meetings and to the period 2009–2015 with most active QE policies.

The full sample gives the general mapping: the FFR factor mixes the policy and Fed response to news shocks, the FG factor mixes all four, and the LSAP factor is mostly the risk premium shock. The policy and news shocks enter the FFR factor together because they are proportional in every asset price and differ only in predictability, so a factor rotation that does not condition on pre-announcement information cannot separate them. The FG factor is the least “structural” of the three: it combines all four shocks, including the risk premium shock, so part of measured forward guidance is premium variation, entering with a negative sign, and part of the measured FFR surprise is the Fed response to news. The LSAP factor, in contrast, is related to a single risk premium shock: no other shock has a comparable coefficient in the LSAP regression in any of the three samples (≈ 0.4), and over 2009–2015 the four shocks explain nearly all of LSAP variation. Thus, the LSAP factor and the risk premium shock are two views of the same variation.

Once unscheduled announcements are excluded, the risk premium loading on the FFR factor loses statistical significance at the 10% level: whatever premium variation the “target” factor carries comes from the intermeeting policy moves. The fit of the FFR regression also falls substantially in both subsamples compared to the full sample. Most of the factor’s variation is not explained by the four shocks, which makes the FFR column the weakest part of the mapping. The FG and LSAP columns remain stable across the subsamples.

Table 10: Swanson (2021) factors regressed on the four JKCS shocks

	All meetings			Scheduled meetings			LSAP period (2009–2015)		
	FFR	FG	LSAP	FFR	FG	LSAP	FFR	FG	LSAP
MP	0.415***	0.332***	0.018	0.225***	0.519***	0.037	−0.026*	0.702***	0.045
FRN	0.327***	0.301***	0.025	0.136***	0.459***	0.045*	−0.000	0.529***	−0.105***
CBI	−0.003	0.495***	0.086*	0.032	0.465***	0.099*	0.015	0.439***	0.012
RP	0.175***	−0.372***	0.394***	0.072	−0.290***	0.398***	−0.039**	−0.142**	0.448***
N		239			223			55	
R^2	0.435	0.637	0.631	0.173	0.738	0.663	0.198	0.834	0.907

Notes. Each column regresses a Swanson (2021) factor on the four JKCS shocks, evaluated at the Fry–Pagan rotation, with same-day statement and press-conference windows summed to one observation per announcement date. Standard errors are HC1 heteroskedasticity-robust. Samples: all FOMC meetings, scheduled meetings only, and the LSAP period January 2009 – December 2015. *** ** *: significance at the 1%, 5%, and 10% levels.

5.5 The shock series

Figures 2a–2d plot each shock as identified by each system that contains it. The series that appear in more than one system track each other closely across three decades, in line with the high correlations, both in normal and in crisis periods. The two risk premium series are almost indistinguishable from each other at every date, including at the largest realizations. The monetary policy shock is the exception, and while the JK and JKCS series are hard to tell apart; the CS series departs from both at the dates where the Fed’s response to news is largest.

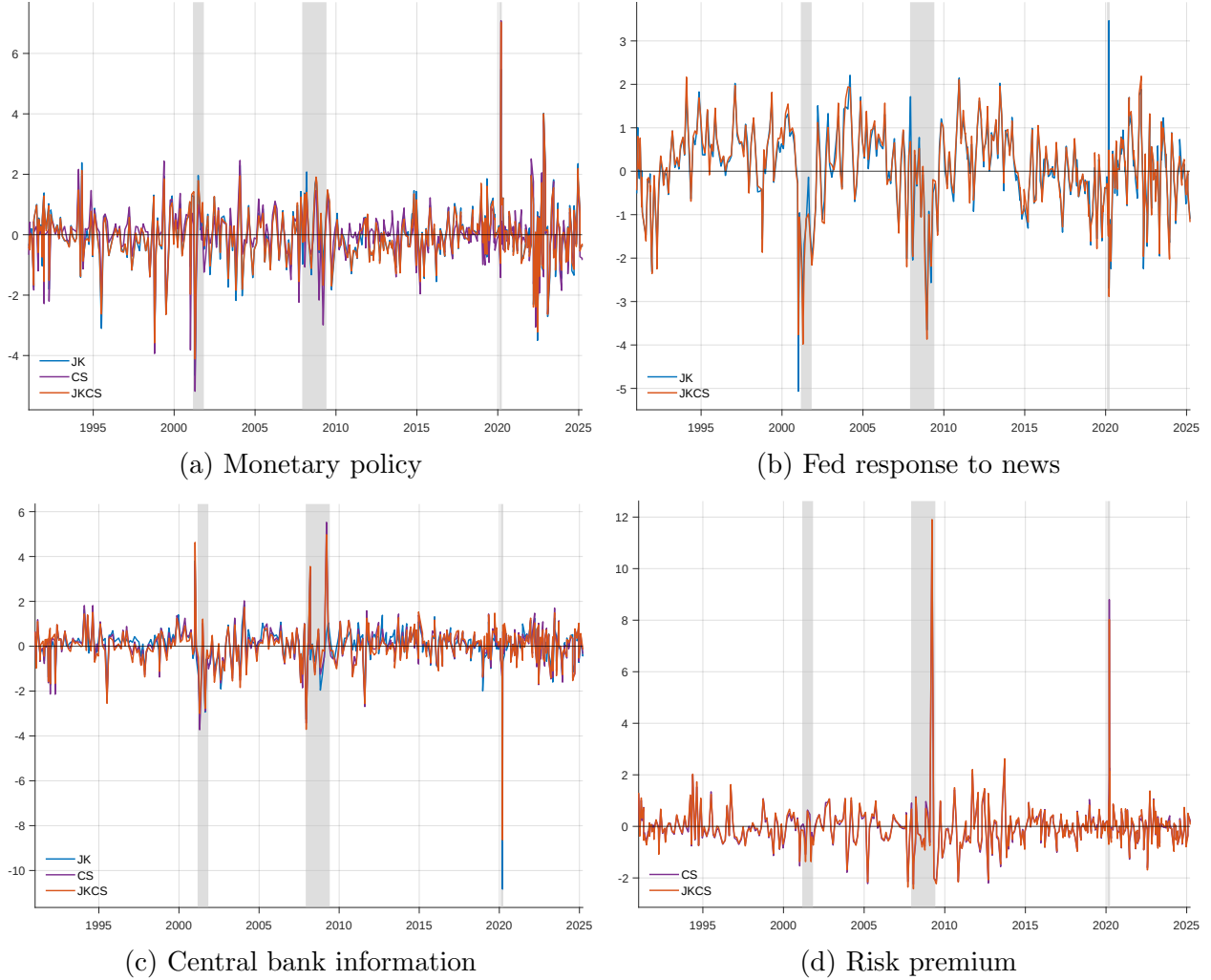


Figure 2: Shock series across identification systems

Notes. Each line is the shock at the Fry–Pagan rotation of its own system; grey areas are crisis periods. $N = 389$ announcement windows, 1991–2025.

5.6 One announcement, three systems

The comparisons above establish how the three systems differ on average. To see the role of the shocks, consider an individual announcement made on 18 September 2007, when the FOMC cut the target rate by 50 basis points, the first cut of the easing cycle, after financial markets started experiencing liquidity shortages. Figure 3 decomposes the window response of the two-year yield, the ten-year yield and the S&P500 into contributions from each shock, one row per system. In the 30-minute window around the announcement the two-year yield fell by about 12 basis points and the S&P500 rose by about 1.4%, which an easing surprise accounts for. The ten-year yield rose by 0.8 basis points.

In the four-shock system (last row) the policy, response-to-news and information shocks

all push both yields down. The announcement combined an unexpected easing of the policy rate, bad news about the economy, and a Fed that responded to that news more aggressively than the market had believed it would. Despite all three forces driving rates in the same direction, the ten-year surprise was positive. The risk premium shock is the only one whose response grows with maturity, by the magnitude restrictions in (14), so the rise in the long yield is attributed to it.

Without a risk premium shock the movement at the long end cannot be explained, because none of the other three shocks produces a long yield that rises while the short yield falls. Jarociński and Karadi (2026) omit that shock and do not have the long yield in the set of observables. Without a response-to-news shock, the policy shock absorbs it: Cieslak and Schrimpf (2019) attribute 1.6 percentage points of the equity response to their monetary policy shock, which roughly corresponds to the sum of MP and FRN ($1.00 + 0.52$) in the four-shock system. What a system estimates the policy surprise to be depends on which other shocks it allows.

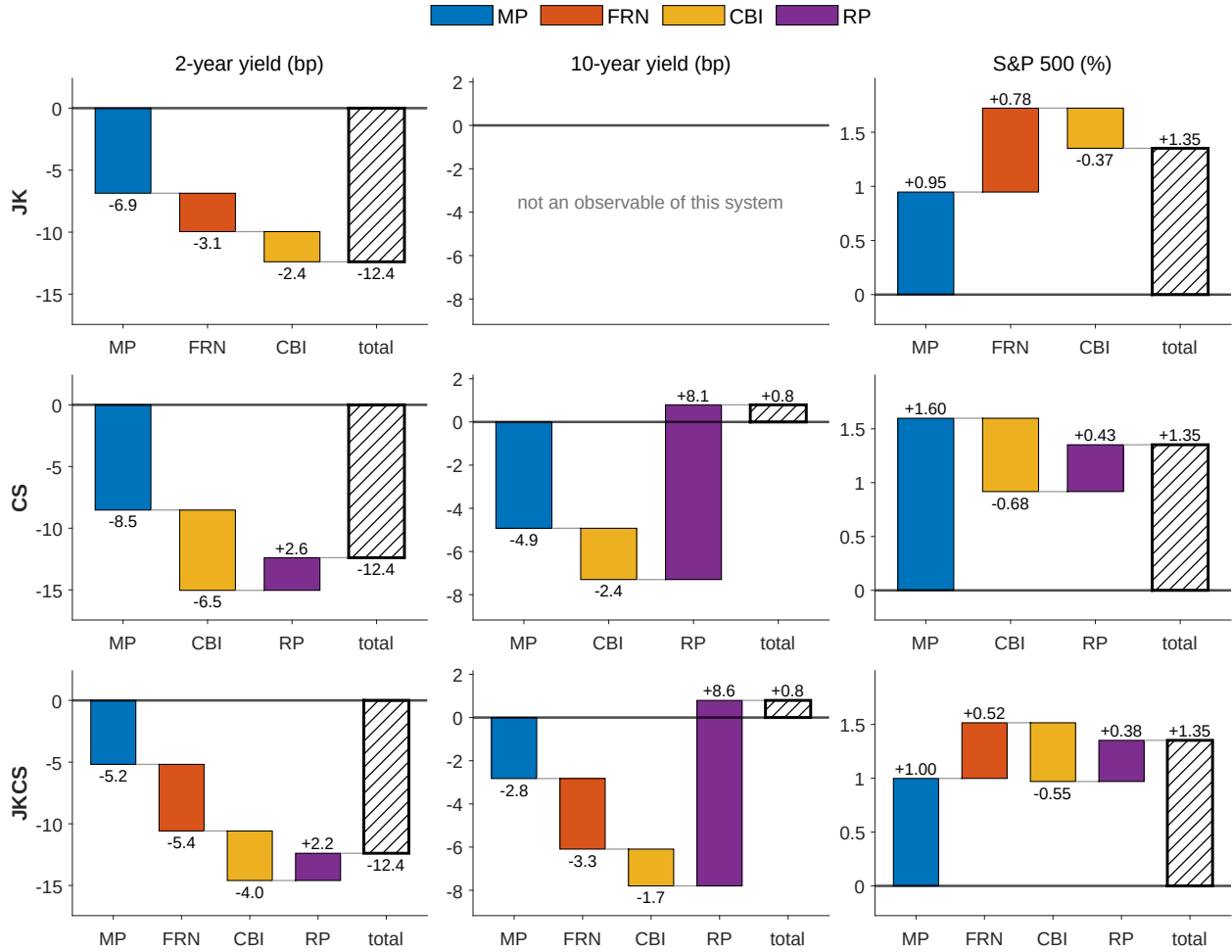


Figure 3: Decomposition of the announcement surprise, 18 September 2007

Notes. Each panel decomposes the surprise in one observable under one system, evaluated at that system's Fry–Pagan rotation. Bars are read left to right: each begins where the previous one ends. The hatched bar is the total actual change.

5.7 The full sample

Figures 4–6 report the decomposition at every announcement in the sample. Each bar splits one window's surprise in the two-year rate, the S&P500 or the ten-year yield into the four shocks, evaluated at the Fry–Pagan rotation, and the four contributions add up to the observed change at that date.

The two-year rate surprise comes from the monetary policy, response-to-news and information shocks, and it does so throughout the sample. The risk premium shock adds little, and what it adds arrives in short episodes rather than across the sample. The equity surprise is shared between the policy and information shocks; the response-to-news and risk premium contributions are small at most announcements.

The risk premium shock is largest in the ten-year yield, where its contribution exceeds that of any other shock. Part of this is built in, since the restrictions require the ten-year yield to respond more to the risk premium shock than to the policy shock. However, no restriction ranks it against the information and response-to-news shocks, and none restricts the size of its contribution or its timing. Appendix 9.3 re-estimates the system with the fourth column of B left free. The signs of its loadings do not change, though the identified set becomes wider.

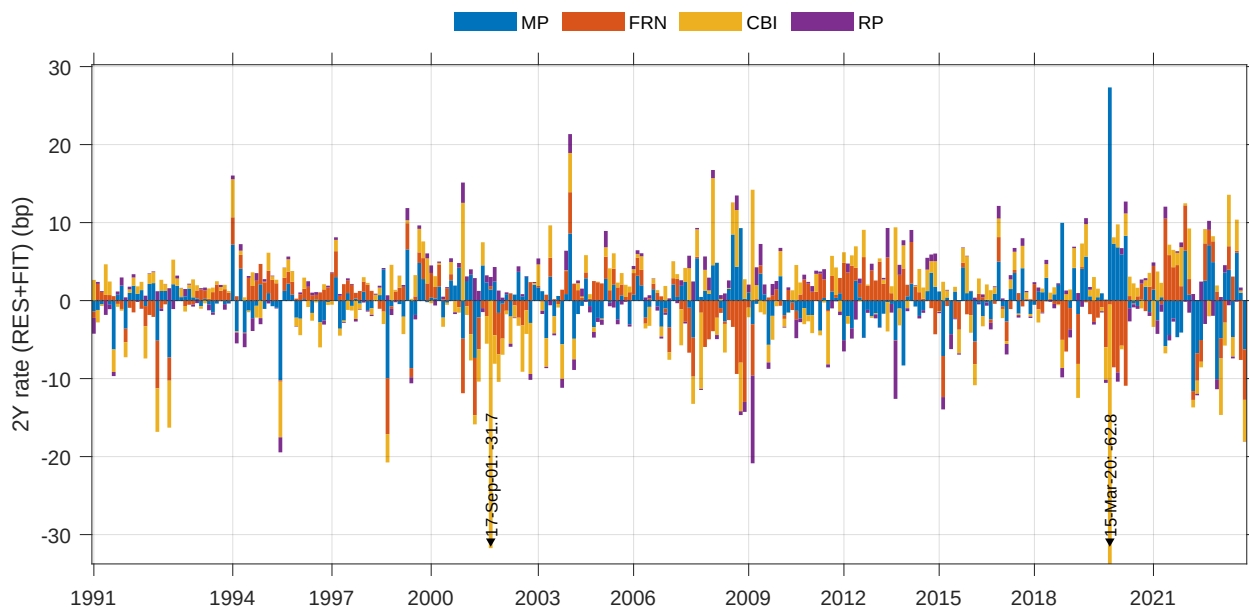


Figure 4: Contributions of the four shocks to the policy surprise, RES + FIT

Notes. Contributions are shown at every announcement, at the Fry–Pagan rotation. The four contributions sum to the observed change. Positive and negative contributions are drawn upward and downward from zero, so the observed change is the height of a bar minus its depth.

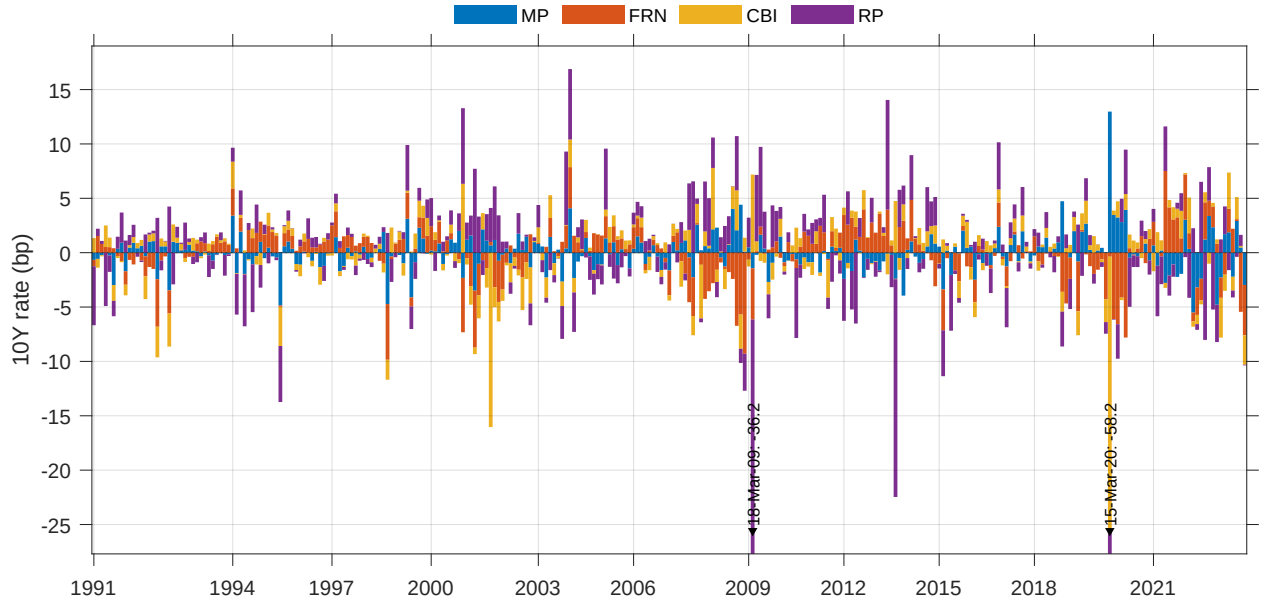


Figure 5: Contributions of the four shocks to the ten-year yield surprise

Notes. Layout as in Figure 4.

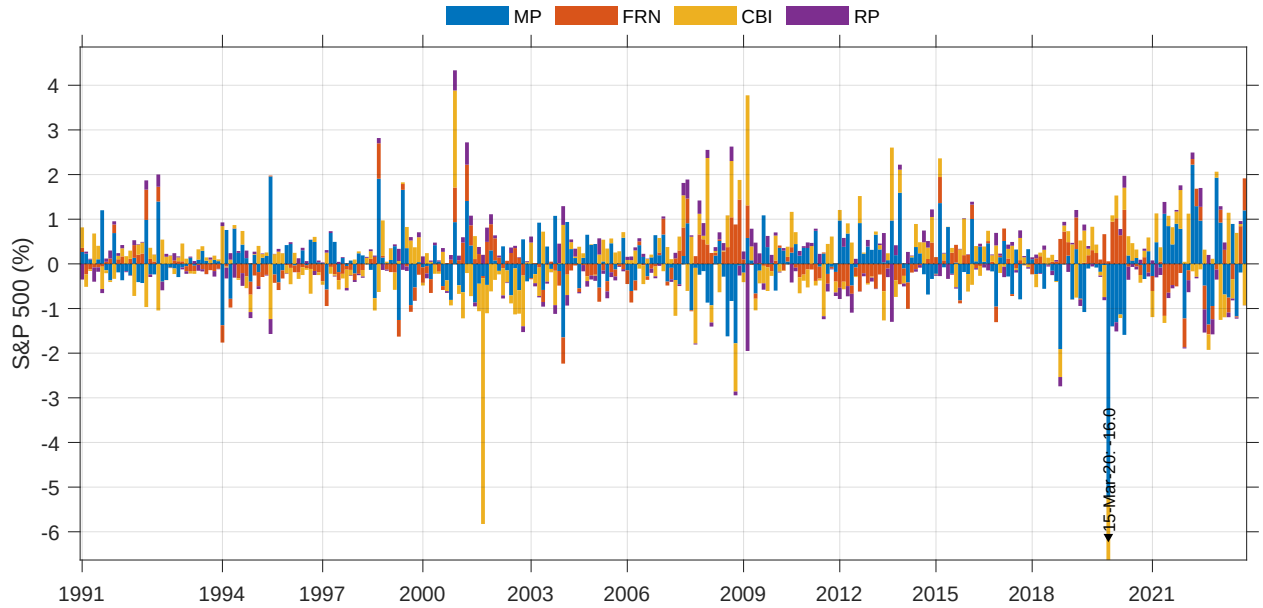


Figure 6: Contributions of the four shocks to the S&P500 surprise

Notes. Layout as in Figure 4.

5.8 Famous announcements

Table 11 reports the decomposition for five FOMC announcements: the expansion of asset purchases of 18 March 2009, the taper communication of 19 June 2013, the decision not

to taper of 18 September 2013, the unscheduled cut to zero of 15 March 2020, which also restarted asset purchases, and the 75 basis point increase of 15 June 2022. For each window it reports the observed change in the ten-year yield and in the S&P500 and each shock's contribution to it, in the same units; the contributions sum to the observed change at every rotation, and the brackets report the $[5, 95]$ range of each across rotations. Three of the announcements concerned the balance sheet, one was a rate decision, and one was both. The identification does not use that distinction: it sees only the window surprises and the restrictions of Section 3. The table asks whether the decomposition separates the two kinds of announcement anyway.

On the three balance-sheet announcements, 18 March 2009, 19 June 2013 and 18 September 2013, the ten-year yield is moved by the risk premium shock. On 18 March 2009 the yield fell by 43 basis points and the decomposition attributes the whole decline to the risk premium shock. On 19 June 2013 it rose by 6.7 basis points in the statement window and 6.0 in the press conference, and on 18 September it fell by 10.6 and 7.0; in all four of those windows the risk premium shock is the largest contributor, and on all three dates its $[5, 95]$ range excludes zero. For 18 March 2009 this agrees with the event-study evidence on QE1, which attributes most of the decline in the ten-year yield around the purchase announcements to the term premium (Sack et al., 2011).

On the rate decision of 15 June 2022 the risk premium shock contributes less than one basis point to the ten-year yield in either window, and the 9.5 basis point decline in the press conference is mostly the policy shock. On 15 March 2020 the ranges are as wide as any in the table and cover zero for the response-to-news shock in both windows and for the risk premium shock in the statement window.

Nothing in the identification distinguishes a balance-sheet announcement from a rate decision, so this alignment is a result of the decomposition; none of the restrictions imposes it. A system without the risk premium shock has to attribute the long-end movement on the balance-sheet dates to the policy or information shock, which is the misattribution Section 5 measures on average.

Table 11: Famous announcements: observed window changes and each shock's contribution.

		Observed	ε^{MP}	ε^{FRN}	ε^{CBI}	ε^{RP}
22-Jan-2008 (unscheduled): <i>Unscheduled intermeeting 75 bp cut</i>						
	ΔR_{10Y} (bp)	+8.8	+2.5 [+0, +5]	-2.7 [-3, -0]	+0.2 [-1, +1]	+8.8 [+6, +11]
	$\Delta S\&P$ (%)	+0.0	-0.9 [-1.2, -0.2]	+0.4 [+0.1, +0.8]	+0.1 [-0.6, +0.2]	+0.4 [+0.0, +0.9]
18-Mar-2009 : <i>QE1 expansion announced</i>						
	ΔR_{10Y} (bp)	-42.9	-3.1 [-14, +3]	-3.2 [-12, +0]	+6.8 [+1, +8]	-43.4 [-48, -29]
	$\Delta S\&P$ (%)	+1.8	+1.1 [-1.6, +4.0]	+0.5 [-0.1, +2.9]	+2.2 [+0.2, +3.7]	-1.9 [-3.7, -0.2]
19-Jun-2013 : <i>Taper communication</i>						
stmt	ΔR_{10Y} (bp)	+6.7	-0.8 [-1, +1]	+2.6 [+0, +5]	-0.5 [-1, +1]	+5.3 [+3, +7]
	$\Delta S\&P$ (%)	-0.0	+0.3 [-0.2, +0.6]	-0.4 [-0.9, -0.1]	-0.2 [-0.3, +0.3]	+0.2 [+0.0, +0.6]
press	ΔR_{10Y} (bp)	+6.0	-0.3 [-1, +2]	+2.9 [+0, +5]	-1.3 [-2, +0]	+4.7 [+2, +6]
	$\Delta S\&P$ (%)	-0.5	+0.1 [-0.5, +0.5]	-0.5 [-1.2, -0.1]	-0.4 [-0.6, +0.1]	+0.2 [+0.0, +0.4]
18-Sep-2013 : <i>No-taper surprise</i>						
stmt	ΔR_{10Y} (bp)	-10.6	-2.3 [-5, -0]	-0.4 [-2, +0]	+1.7 [+0, +2]	-9.6 [-11, -5]
	$\Delta S\&P$ (%)	+1.0	+0.8 [+0.0, +1.5]	+0.1 [-0.1, +0.7]	+0.6 [+0.0, +1.1]	-0.4 [-0.7, -0.1]
press	ΔR_{10Y} (bp)	-7.0	-0.7 [-2, +0]	+0.1 [-1, +0]	+1.1 [+0, +1]	-7.5 [-8, -5]
	$\Delta S\&P$ (%)	+0.3	+0.3 [-0.2, +0.7]	-0.0 [-0.1, +0.3]	+0.4 [+0.0, +0.6]	-0.3 [-0.6, -0.0]
15-Mar-2020 (unscheduled): <i>Unscheduled cut to the zero lower bound</i>						
stmt	ΔR_{10Y} (bp)	-0.8	+12.8 [+4, +18]	+0.7 [-3, +8]	-9.8 [-14, -1]	-4.5 [-17, +2]
	$\Delta S\&P$ (%)	-8.0	-4.5 [-7.1, -1.2]	-0.1 [-3.5, +0.1]	-3.1 [-6.2, -0.2]	-0.2 [-2.1, +0.0]
press	ΔR_{10Y} (bp)	-32.5	+8.1 [+1, +11]	+0.4 [-7, +6]	-11.7 [-20, -1]	-29.3 [-41, -21]
	$\Delta S\&P$ (%)	-8.0	-2.9 [-6.0, -0.1]	-0.1 [-3.4, +0.3]	-3.8 [-6.3, -0.3]	-1.3 [-4.8, -0.1]
15-Jun-2022 : <i>75 bp hike, first since 1994</i>						
stmt	ΔR_{10Y} (bp)	+2.0	+1.1 [+0, +2]	+0.6 [-0, +1]	+1.3 [+0, +3]	-0.9 [-1, +1]
	$\Delta S\&P$ (%)	-0.1	-0.4 [-0.5, -0.1]	-0.1 [-0.2, +0.0]	+0.4 [+0.1, +0.4]	-0.0 [-0.1, +0.0]
press	ΔR_{10Y} (bp)	-9.5	-5.9 [-9, -2]	-0.8 [-2, +0]	-2.3 [-6, -0]	-0.5 [-5, +1]
	$\Delta S\&P$ (%)	+1.5	+2.1 [+1.3, +2.2]	+0.1 [-0.1, +0.5]	-0.7 [-0.7, -0.0]	-0.0 [-0.2, +0.1]

Notes: Each row is a single announcement window, so every shock is one unit-variance draw; where a date carries both a statement and a press conference the two are shown separately. The shocks are identified on the pooled window set in every case. Each row decomposes the observed change of the named variable into the shocks' contributions, in the same units; the contributions sum to the observed change exactly, at every accepted rotation. Contributions are evaluated at the Fry–Pagan rotation; the figure beside each is the [5,95] range of that contribution across accepted rotations. The observed change is data and does not vary with the rotation; only its split does. Changes are net of the sample mean of the announcement-window changes, -0.50 bp for the ten-year yield and -0.02% for the S&P500. The shocks respond to window surprises, not to the policy action: a cut smaller than its predictable component is a positive policy shock.

5.9 Robustness

Table 12 reports the variance decomposition of the four-shock system in seven configurations: the baseline and six variants, each changing one element of the identification or of the event set. Panel A reports the risk premium share of each observable and Panel B the full decomposition of the ten-year yield. Rows (2) and (3) address the external restriction, which identifies the risk premium shock in part through its correlation with an outside index of risk appetite: row (2) replaces the Bauer et al. (2023) index with the Bekaert et al. (2022) risk-aversion index and row (3) drops the external restriction. Row (4) drops the restriction

in (15) that orders the risk premium shock above the policy shock at the ten-year yield. Rows (5) to (7) change the event set: they remove the unscheduled announcements, keep only the statement windows, and add the minutes releases.

Overall, the decomposition results remain robust in all of considered configurations. The risk premium shock is the largest contributor to ten-year variance in all of them, with a share between 61.8 and 71.5%, and at its smallest it is still more than three times any other shock's share. Replacing the index used for external restriction leaves every point estimate unchanged. Likewise, row (4) reproduces the baseline point estimates, and at the center of the identified set this restriction is nearly redundant. Rows (1), (2) and (4) share the event set and therefore the reduced-form covariance matrix; the restrictions that differ between them do not bind at the Fry–Pagan rotation, so the point estimates coincide and only the bounds of the identified set move slightly.

Appendix 9.3 leaves the fourth column of B free of any sign, magnitude, or external restriction. The shock that comes out still lowers equity prices and the unpredictable part of the policy surprise, and the other three shocks are almost unchanged. What the restrictions buy is precision: the 5-95% band on the risk premium share of ten-year variance widens from 36.9 to 76.0 percentage points without them.

Table 12: Robustness of the four-shock decomposition

Configuration	RES	FIT	$\Delta S\&P500$	ΔR_{10Y}
<i>Panel A. Share of each variable attributed to the risk-premium shock (%)</i>				
(1) Baseline (BBM ext. rest.)	2.7 [0.3, 22.1]	0.4 [0.0, 9.8]	3.8 [0.0, 26.4]	64.2 [47.1, 84.0]
(2) BEX instead of BBM	2.7 [0.2, 23.4]	0.4 [0.0, 6.4]	3.8 [0.0, 25.0]	64.2 [45.6, 80.6]
(3) No ext. rest.	8.7 [0.3, 23.0]	0.0 [0.0, 15.5]	6.0 [0.0, 25.3]	71.5 [47.7, 87.9]
(4) – RP moves R_{10Y} more than MP	2.7 [0.3, 21.9]	0.4 [0.0, 9.8]	3.8 [0.0, 28.9]	64.2 [45.7, 84.0]
(5) Scheduled meetings only	4.4 [0.2, 11.6]	3.8 [0.0, 17.7]	0.7 [0.0, 9.9]	69.7 [47.8, 76.7]
(6) Statement windows only	6.0 [0.3, 16.3]	0.1 [0.0, 10.1]	4.8 [0.0, 13.7]	61.8 [43.8, 75.9]
(7) Minutes releases added	2.7 [0.2, 18.8]	0.4 [0.0, 8.9]	4.0 [0.0, 26.4]	64.0 [49.1, 80.1]
<i>Panel B. Full decomposition of ΔR_{10Y} (%)</i>				
	MP	FRN	CBI	RP
(1) Baseline (BBM ext. rest.)	16.1 [1.7, 35.6]	10.7 [0.3, 29.0]	9.0 [0.1, 27.9]	64.2 [47.1, 84.0]
(2) BEX ext. rest. instead of BBM	16.1 [1.3, 34.5]	10.7 [1.3, 30.3]	9.0 [0.1, 30.2]	64.2 [45.6, 80.6]
(3) No ext. rest.	13.6 [1.9, 35.3]	11.3 [0.1, 28.2]	3.6 [0.0, 26.4]	71.5 [47.7, 87.9]
(4) – RP moves R_{10Y} more than MP	16.1 [1.8, 37.2]	10.7 [0.3, 29.1]	9.0 [0.0, 27.6]	64.2 [45.7, 84.0]
(5) Scheduled meetings only	21.1 [12.2, 41.3]	7.1 [0.1, 24.3]	2.1 [0.0, 10.8]	69.7 [47.8, 76.7]
(6) Statement windows only	19.1 [4.2, 39.0]	17.0 [0.8, 36.5]	2.1 [0.0, 18.3]	61.8 [43.8, 75.9]
(7) Minutes releases added	18.6 [3.1, 39.7]	8.5 [0.1, 19.3]	8.9 [0.1, 30.0]	64.0 [49.1, 80.1]

Notes. Rows (5) to (7) change the event set and so do not share the baseline's $\hat{\Sigma}$. Points and brackets follow Table 5; Panel B rows sum to 100 and bold marks the largest entry.

6 Impact of shocks on financial and macroeconomic indicators

6.1 Financial variables

To see how the shocks affect financial indicators, for each indicator y and each horizon h I estimate the local projection

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{cumulative change}} = a_h + b'_h \varepsilon_t + u_{t,h}, \quad h = 0, 1, \dots, 30, \quad (17)$$

where t runs over FOMC announcement days. The dependent variable is the cumulative change in y from the end of the day before the announcement to the end of the day h business days after it, so $h = 0$ is the announcement-day response and $h = 30$ is the change over the following six weeks. The regressor ε_t collects the four structural shocks; on days with two announcement windows the shocks are summed. Each shock has unit variance, so b_h is the response to a one-standard-deviation shock in the units of y .

The regressors ε_t are estimated, and each accepted rotation satisfies all restrictions equally well. Different rotations produce different shock series and therefore different coefficients in the local projection. I summarize the uncertainty by a single distribution that combines rotation uncertainty across the identified set with estimation uncertainty in the local projections. To do that, I take the shocks from one accepted draw of the identification stage, estimate the local projection with heteroskedasticity- and autocorrelation-consistent standard errors, and repeat this for all $K = 2000$ accepted draws, collecting $\{\hat{b}_h^{(k)}, \widehat{\text{se}}(\hat{b}_h^{(k)})\}_{k=1}^K$. For each draw I then take $M = 20$ values from $N(\hat{b}_h^{(k)}, \widehat{\text{se}}(\hat{b}_h^{(k)})^2)$, the normal approximation to the sampling distribution of the coefficient at that rotation, and pool the $K \times M$ values. The 90 and 68 percent bands are the 5th and 95th and the 16th and 84th percentiles of the pooled values, computed separately for each variable, shock, and horizon; the point estimate is the coefficient at the Fry–Pagan draw.

Figure 7 reports the responses of six financial indicators, one per row, to the four shocks, one per column: 2-year Treasury yield, the term premium and the expectations component of the ten-year Treasury yield from the decomposition of Adrian et al. (2013), the S&P 500, the ICE BofA BBB corporate spread, and the risk appetite index of Bauer et al. (2023).

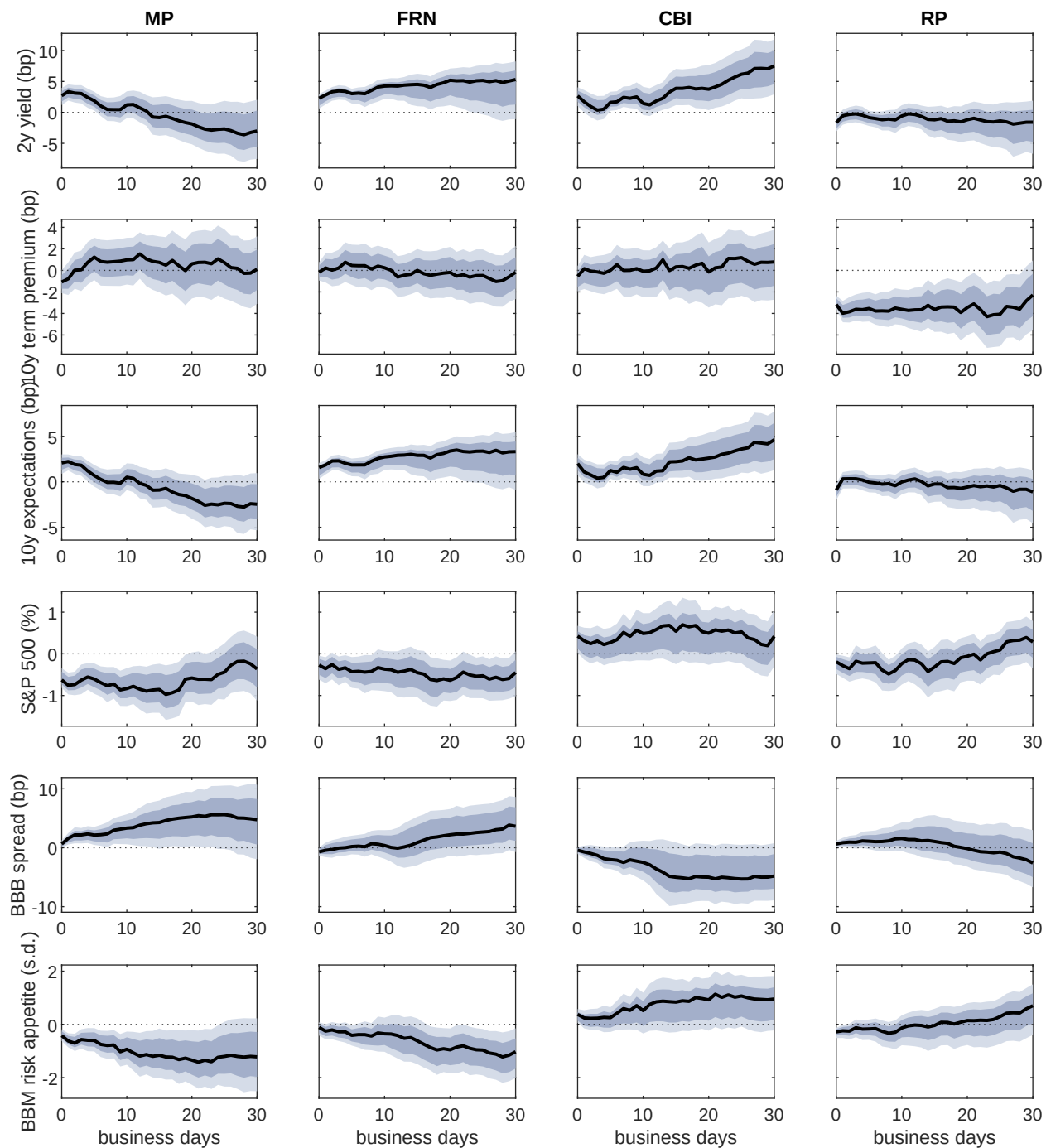


Figure 7: Daily responses to the four shocks

Notes. Solid lines are point estimates at the Fry–Pagan rotation; the light and dark shaded bands are the 90% and 68% ranges of the combined distribution described in the text. The discussion refers to the 90% band.

In each column of Figure 7 the responses of most of financial indicators have the signs the shock’s label implies. The monetary policy and Fed-response-to-news shocks have standard contractionary effects on daily data: raise the two-year yield on impact and lower equity

prices; over the following weeks both widen the BBB spread by about 5 basis points and reduce measured risk appetite by about one standard deviation. The two-year yield response to FRN rises to about 5 basis points and stays there through the end of the horizon; the response to MP returns to zero within about two weeks. The central bank information shock raises the two-year yield as well, to about 8 basis points by the end of the horizon, and results in higher equity prices.

The risk premium shock lowers the ten-year term premium by about 3 basis points on impact, and the response stays 2 to 4 basis points below zero at every horizon, while the two-year yield and the expectations component of the ten-year move by about 2 basis points or less, with bands covering zero after the announcement day. The other three shocks, in contrast, do not affect the term premium significantly, and seem to operate more through the expectations part of the yield curve. The RP shock therefore moves the long end through the compensation investors require for bearing duration risk and not through the expected path of policy. This feature is not built into the identification: the restrictions require the risk premium shock to lower the ten-year yield, but they do not say which part of the yield moves. That is left to the data, and the data attribute the decline to the term premium.

The risk premium shock has small effects on equities, credit spreads and measured risk appetite, and their bands cover zero at almost every horizon. The label “risk premium” leads one to expect credit spreads and risk appetite to move; they do not, because the shock is, essentially, a term premium shock. The restrictions identify it through the maturity structure of the yield response and leave its effect on risky assets free beyond the sign, and what they isolate is the part of the long-yield move that the expected policy path does not explain. A term premium is the compensation for bearing interest-rate risk, which Treasury and corporate bonds share: a change in it moves both yields together and leaves the spread unchanged.

Appendix 9.5 reports the same projections for the full set of variables, with a robustness specification that adds five lags of the daily change in the dependent variable. The two lines are close in most panels, and the signs of the responses remain unchanged.

6.2 Macroeconomic variables

To see how the shocks effect macroeconomic variables at monthly frequency, I estimate a monthly Bayesian VAR in the style of Jarociński and Karadi (2026). The VAR contains four shocks, ordered first, followed by five monthly variables: the one-year Treasury yield, the log S&P500, the excess bond premium of Gilchrist and Zakrajšek (2012), log real GDP, and the log PCE price index. Section 4.3 describes their construction and sources. The shocks

enter as within-month sums of the high-frequency shocks, with zero in months without an announcement, and the March 2020 announcements are excluded, following Jarociński and Karadi (2026). Identification is recursive with the shocks ordered first, so the shock columns do not depend on how the macro variables are ordered among themselves. The VAR has 12 lags and a Minnesota prior with overall tightness 0.2, estimated on 1990:05–2025:03. Figure 8 reports responses to one-standard-deviation shocks. Similar to daily local projections, the constructed credible bands carry both parameter and identification uncertainty over 2000 accepted draws. The solid line corresponds to the center of the identified set in the sense of Fry and Pagan (2011); a pointwise median across rotations is not itself an admissible impulse response.

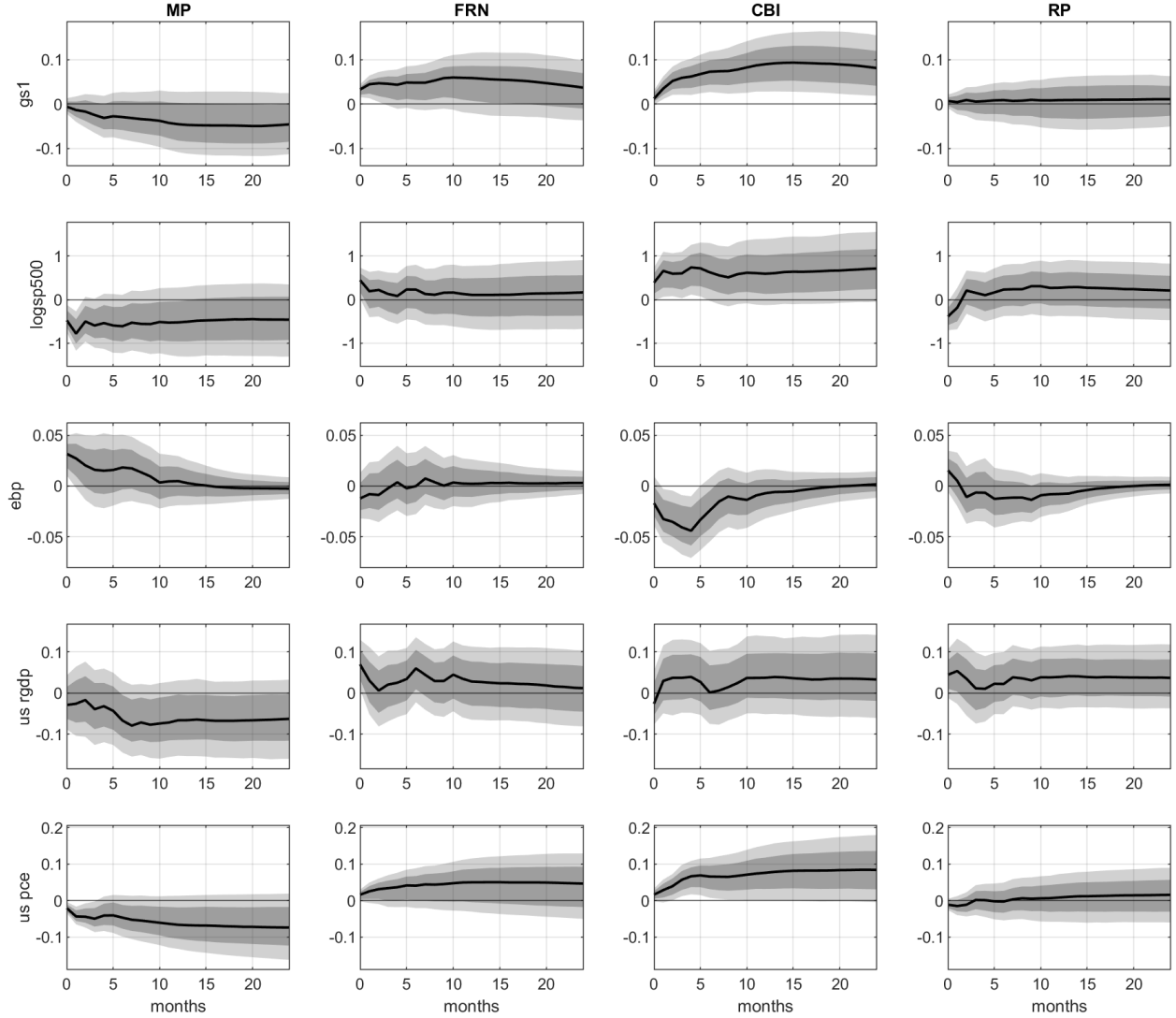


Figure 8: Monthly BVAR impulse responses to the four shocks

Notes. Columns: monetary policy (MP), Fed response to news (FRN), central bank information (CBI), and risk premium (RP) shocks; rows: the one-year Treasury yield (gs1), log S&P500 (logsp500), excess bond premium (ebp), log real GDP (us rgdp), and log PCE price index (us pce). Solid lines are posterior medians at the Fry–Pagan rotation; shaded areas are 68 and 90 % posterior bands from 2,000 draws over the accepted rotations.

The estimated impulse responses are largely consistent with macroeconomic theory and existing empirical evidence, with one exception, the one-year yield response to the monetary policy shock, which Section 6.4 discusses separately. Apart from that, a contractionary monetary policy shock produces the standard effects: stock prices fall, the excess bond premium rises, real GDP declines, and the price level falls persistently, with no price puzzle. The one-year yield itself moves little on impact and drifts mildly negative at longer horizons.

A positive response-to-news shock, a stronger response to public news than the market

expected, raises the one-year yield gradually and persistently, in line with a systematic policy change. Elsewhere it has expansionary effects: stock prices rise, the excess bond premium falls, and activity and the price level increase, a temporary boom much like the one that follows an information shock. Since growth is persistent, the good news that prompted rate increase today, keeps arriving for months, and the monthly responses record that expansionary effect from news rather than the policy move. A rate increase followed by a boom is what an econometrician would attribute to favorable information held by the Fed, which is why the response resembles the one to the information shock (Bauer and Swanson, 2023a; Jarociński and Karadi, 2026). A rate increase that is caused by the Fed’s response to good public news therefore has the opposite effects of a policy shock: stock prices, activity and the price level rise. Jarociński and Karadi (2026), in contrast, found insignificant effects of FRN on economic activity, consistent with the stimulative effects of favorable fresh public news and the contractionary effects of tighter than expected systematic monetary policy broadly offsetting each other.

The risk premium shock is a financial repricing with little macroeconomic follow-through. A risk-off shock lowers stock prices and raises the excess bond premium only on impact; both effects reverse within a few months, and the responses of activity and prices are not statistically significant. Unlike a policy tightening, the premium shock moves asset prices without a persistent real or nominal adjustment. Finally, although the responses are consistent with theory overall, they are estimated with wider bands than in most earlier studies.

6.3 Comparison with the JK and CS systems

Figures 9 and 10 place the JKCS responses on top of the JK and CS bands for the shocks each pair shares. First, adding the risk premium shock to the JK system leaves the monthly responses of the other three shocks essentially unchanged. At monthly frequency, omitting the risk premium shock changes nothing about the other three shocks that the bands can detect.

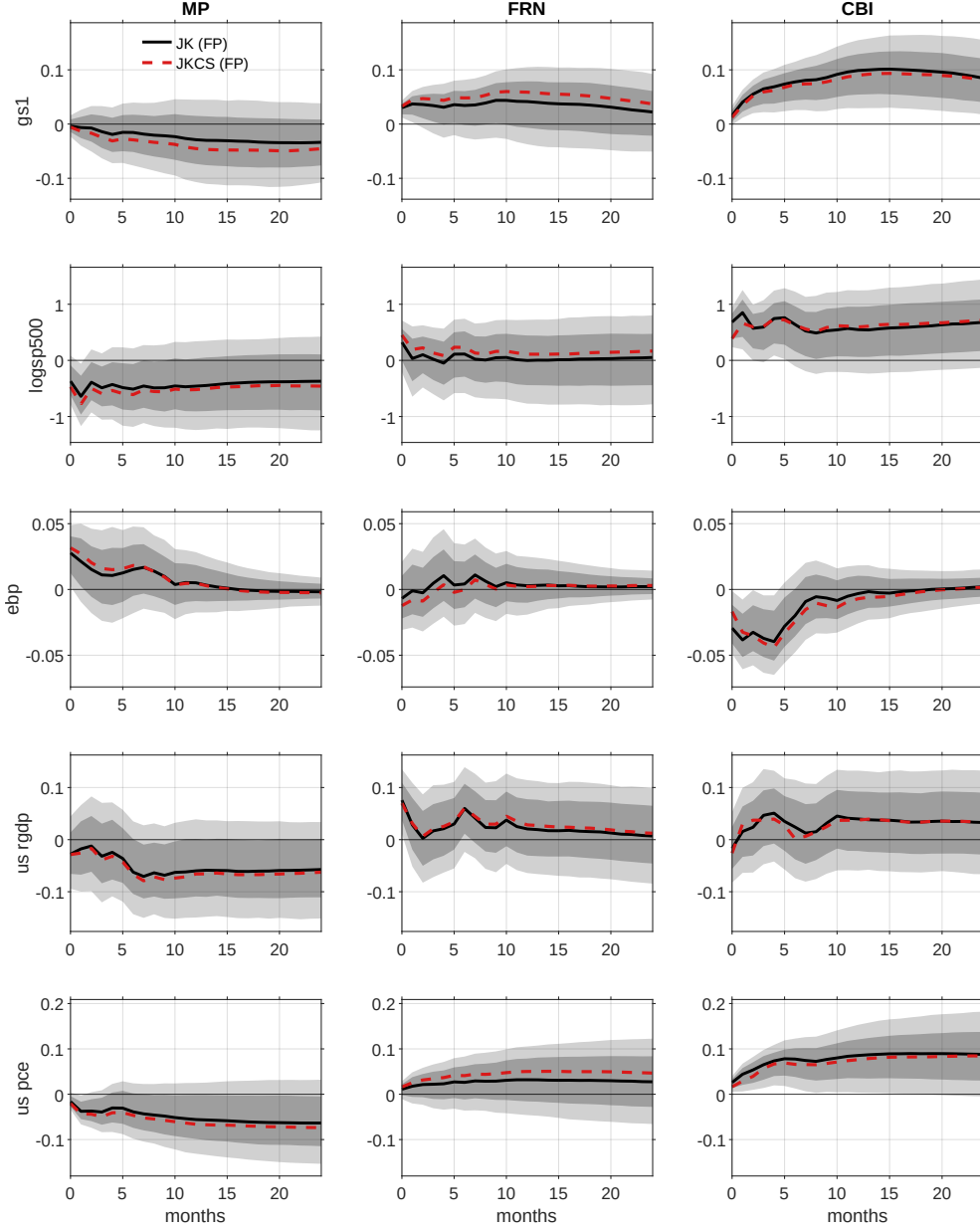


Figure 9: Monthly responses to the three shocks JK and JKCS share

Notes. Black solid lines and shaded areas are the JK system: the posterior median at the Fry–Pagan rotation, with 68 and 90% posterior bands over all accepted rotations. Red dashed lines are the JKCS system at its own Fry–Pagan rotation. Neither line is rescaled; each is a one-standard-deviation shock in its own units.

The CS and JKCS systems agree on CBI and RP shocks effects and differ on MP. In the MP column the JKCS responses of the S&P500, real GDP and the price level are 1.5-2 times as large as the CS responses, and the one-year yield response is the more negative of the two at longer horizons. This is what Table 7 implies. A fifth of the CS policy shock is the response-to-news shock, whose monthly responses are expansionary, so the CS policy shock

carries a component that offsets part of its contractionary effect. Omitting the response-to-news shock underestimates the effects of monetary policy. The monthly VAR does not, however, establish that the two responses are distinguishable: in every row of the MP column the JKCS line lies inside the CS 90% band.

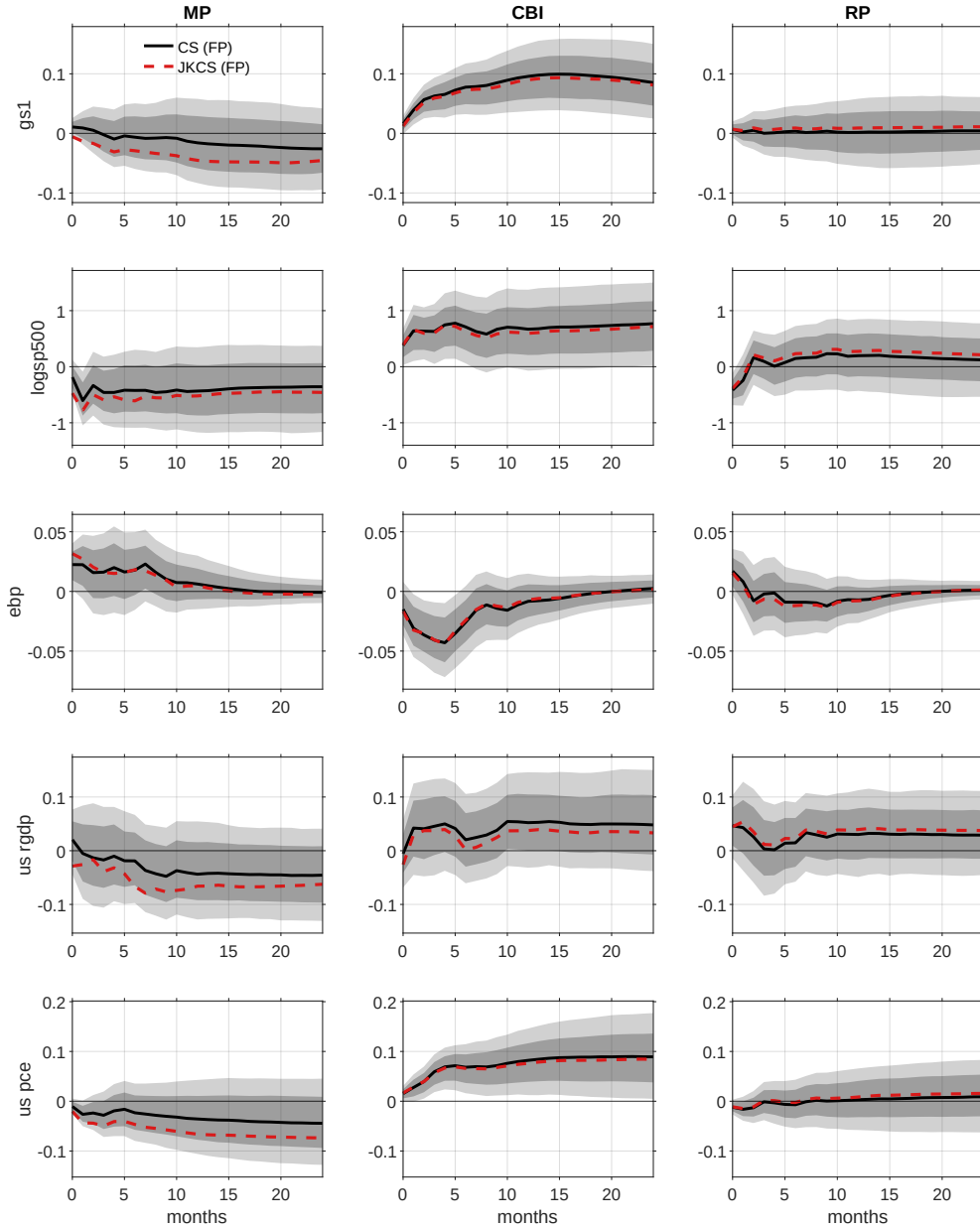


Figure 10: Monthly responses to the three shocks CS and JKCS share

Notes. Black solid lines and shaded areas are the JK system: the posterior median at the Fry–Pagan rotation, with 68 and 90% posterior bands over all accepted rotations. Red dashed lines are the JKCS system at its own Fry–Pagan rotation. Neither line is rescaled; each is a one-standard-deviation shock in its own units.

6.4 The one-year yield response to the monetary policy shock

The one-year yield response to the monetary policy shock in Figure 8 is negative at the point estimate at every horizon, including impact, and its band includes zero throughout, while every other row of the MP column shows the standard contractionary effect of the shock. Such puzzling response is not specific to this system: Figures 9 and 10 show the same shape for JK and for CS on the same announcements. It is also visible in the daily local projections: despite a positive effect on the yield in the first couple of weeks, the reaction eventually becomes insignificant.

The same feature discussed in Jarociński and Karadi (2026), and their estimates locate it. On FOMC announcements alone, with sign restrictions only, which is the design of this paper, their monetary policy shock does not move the one-year yield. In their baseline, which has a larger event set (speeches, testimonies and minutes releases) and additional heteroskedasticity-based restrictions, the reaction of the one-year rate is conventional. Thus, the reason for the puzzling reaction lies either in the event set or in the heteroskedasticity-based part of the identification. Its investigation is left for future work.

7 Discussion

This section discusses the implications of the results for the choice of identification.

The monetary policy shock. The Jarocinski–Karadi three-shock identification is sufficient for the monetary policy shock identification: adding the risk premium shock leaves its policy shock almost unchanged, and the monthly responses of the two policy shocks lie inside each other’s bands. A three-shock system without the response-to-news shock is not sufficient, however. A sizable part of the Cieslak–Schrimpf policy shock is the Fed’s response to news, and in the monthly VAR it makes the responses of equities, activity and prices to the policy shock noticeably smaller than their four-shock size.

The information shock. The choice among the identifications considered here matters little for the CBI shock. Under the Jarocinski–Karadi restrictions the information shock absorbs small parts of the risk premium and response-to-news shocks, but its correlation with the four-shock counterpart is high. Under the Cieslak–Schrimpf restrictions, which include a risk premium shock, it is almost unchanged. The main conclusion is that the central bank information effect survives the addition of the risk premium shock; the part of it that is risk-premium variation is small and does not change the estimated effects of the shock.

The composition of the announcement surprise. If the object of interest is what moves markets in the announcement window, the four-shock system is necessary. The risk premium shock is the largest single contributor to the ten-year surprise, in the baseline and in every configuration considered; it accounts for most of the ten-year move on most FOMC dates; and in daily projections it lowers the ten-year yield through the term premium and not through the expected path of policy. A system without it does not lose that variation, it relabels it: the Jarocinski–Karadi system does not observe the ten-year yield, and the risk-premium variation in the surprises it does observe is absorbed by its CBI shock, so a small part of what that system calls information is risk premium.

The risk premium shock. The estimate of the risk premium shock does not depend on whether the Fed-response-to-news shock is accounted for. The information the four-shock system adds (the decomposition of the monetary policy surprise into its predictable and unpredictable parts) is nearly orthogonal to the risk premium shock, which accounts for very little of either part. As identified here, the shock is primarily a term-structure shock: its share of the equity surprise is small and imprecisely estimated, and its daily effects on credit spreads and risk appetite are not distinguishable from zero. For a researcher interested in this shock, it should not matter whether the Fed-response-to-news shock is accounted for; the interpretation of the shock, however, is narrower than the one Cieslak and Schrimpf (2019) give it.

Limitations. These conclusions come with three limitations. First, the identification is set identification from sign and magnitude restrictions in the announcement window, and some of the objects it delivers are imprecise. Jarociński and Karadi (2026) sharpen the same kind of restrictions with the difference in shock variances between FOMC and non-FOMC events, which is not available in a sample of FOMC announcements. Second, the response-to-news shock is defined relative to a public information set. While the set used here is the union of indicators from several papers, the choice of regressors for the monetary policy surprise matters for the results (Laumer, 2026). Third, the event set is FOMC statements and press conferences only. Jarociński and Karadi (2026) add Chair speeches, testimony and minutes releases and find that the mix of shocks differs across event types; whether the risk premium shock behaves the same way at those events is an open question.

8 Conclusion

This paper proposes a high-frequency identification that recovers four shocks jointly at FOMC announcements: monetary policy, central bank information, the Fed’s response to news, and the risk premium. The identification is based on sign and magnitude restrictions on how each shock moves bond and stock prices, derived from a small asset pricing model of the announcement window in the spirit of Cieslak and Schrimpf (2019). The key feature of the model is the Fed response to news shock: the market underestimates how strongly the Fed reacts to growth, making part of the policy surprise predictable from pre-announcement publicly available information. Using the model predictions, I embed the restrictions in a simple factor model, estimate all four structural shocks on the same sample of announcement windows from 1991 to 2025, and compare them with the three-shock systems of Jarociński and Karadi (2026) and Cieslak and Schrimpf (2019) to see what happens when one of the shocks is not identified.

Three findings stand out. First, the risk premium shock accounts for more than 60% of the variance of the ten-year yield change in the announcement window, and for less than 5% of the changes in the two-year yield and the stock market. In the weeks after an announcement it lowers the ten-year yield through the term premium and not through expected policy rates, while its effects on stock prices, corporate credit spreads and risk appetite are small. As identified in the paper, the risk premium shock is a term premium shock, a narrower object than its label suggests, and at monthly frequency it moves asset prices without any effect on activity or prices.

Second, each omission contaminates a different shock. Omitting the Fed’s response to news, as in Cieslak and Schrimpf (2019), leaves the risk premium shock intact, but contaminates the policy shock: a fifth of their policy shock is actually attributed to the Fed response to news, and the monthly responses of stock prices, activity and the price level to the four-shock policy shock are 1.5–2 times as large as the responses to theirs, though the estimates are imprecise. Omitting the risk premium shock, as in Jarociński and Karadi (2026), leaves the policy and response-to-news shocks almost unchanged. At the same time, 80% of their information shock is the jointly identified information shock and 9% is the risk premium shock, so part of what that system calls the information shock is, in fact, risk premium variation, though the part is qualitatively small.

Third, for applied work the three-shock identification of Jarociński and Karadi (2026) is sufficient for the monetary policy and information shocks: a fifth of its information shock is other shocks, mostly the risk premium, but the estimated monthly responses to the two information shocks are almost identical. A policy shock identified without the Fed’s response

to news, in contrast, mixes monetary policy with the Fed’s predictable response to news and underestimates the effects of policy. To say which shock moves which asset price in the announcement window, the four-shock system is likely necessary.

Two extensions follow. The first is expansion of the event set: adding speeches, testimony and minutes releases makes the heteroskedasticity-based restrictions of Jarociński and Karadi (2026) available, which could help to narrow the identified set. The second is looking at other central banks, to which the model and the restrictions should be equally applicable so announcements of other central banks could serve as external validation of the identification.

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9 Appendix

9.1 Implementation of the restrictions

Let u_t be the $n \times 1$ vector of observables (high-frequency surprises) of a given system, demeaned over the T announcement windows, and let $\hat{\Sigma} = (T - 1)^{-1} \sum_t u_t u_t' = PP'$ be its sample covariance matrix of the observables, with P the lower triangular Cholesky factor. Any orthogonal Q gives an observationally equivalent decomposition,

$$u_t = B \varepsilon_t, \quad B = PQ', \quad \varepsilon_t = B^{-1}u_t, \quad (18)$$

since $BB' = PQ'QP' = \hat{\Sigma}$ whenever $QQ' = Q'Q = I$. Identification is the choice of Q . The restrictions of Section 3 cannot pin down a single Q , a set of rotations satisfy them, which is sampled by the rejection algorithm below.

1. Draw an $n \times n$ matrix M with independent standard normal entries and take its QR decomposition, $M = QR$. If the i th diagonal entry of R is negative, replace the i th column of Q by its negative; Q is then uniform over the orthogonal matrices (Rubio-Ramirez et al., 2010). Form $B = PQ'$.
2. Normalize the sign of each column, for JKCS by the four diagonal entries of (11): $c_{11} > 0$, $c_{22} > 0$, $c_{33} > 0$, $c_{44} < 0$. A column and its negative describe the same decomposition, so this names the shocks and does not restrict the set.
3. Check the sign and magnitude restrictions, (11)–(15) for JKCS, (4) and (5) for JK, and (6)–(10) for CS, and discard at the first failure.
4. For JKCS, form $\varepsilon_t = B^{-1}u_t$ and keep the draw only if the fourth shock satisfies (16). JK and CS impose no external restriction.
5. Repeat steps (1)–(5), storing every accepted B , until 2,000 draws are collected.

Point estimates for empirical exercises, if not mentioned otherwise, are taken at one accepted draw, the rotation of Fry and Pagan (2011),

$$k^* = \arg \min_{k \leq K} \sum_{m=1}^n \sum_{j=1}^n \left(\frac{B_{mj}^{(k)} - \text{med}_i B_{mj}^{(i)}}{\text{sd}_i B_{mj}^{(i)}} \right)^2, \quad (19)$$

where K is the number of accepted draws and the median and the standard deviation are taken over them entry by entry.

9.2 Risk premium shocks: comparison

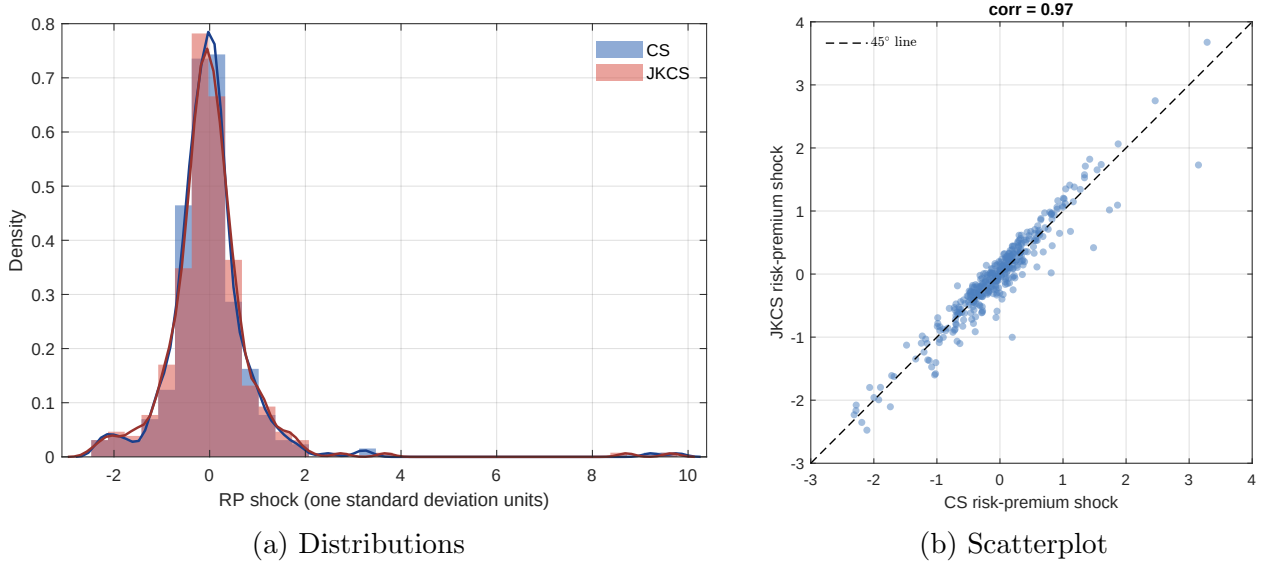


Figure 11: The risk-premium shock under the CS and the four-shock restrictions

Notes. Both series are at the Fry–Pagan rotation of their own system, on the same 389 announcement windows. Panel (a) plots the two distributions and panel (b) the two series against each other.

9.3 How much of the risk premium shock is delivered by the restrictions?

Every restriction on the fourth column of B is a choice, so the risk premium shock could in principle be a creature of those choices rather than of the data. This appendix asks how much of it the restrictions deliver. The exercise estimates the same 4×4 system twice on the four observables of equation (11). The baseline imposes every restriction the model of Section 2 implies. The second estimation keeps all restrictions in the first three columns of B^{JKCS} and leaves the fourth column completely free: no sign, magnitude, or external correlation restriction, except the sign of the ten-year loading, which is the normalization choice. Therefore, the baseline identified set is by construction a subset of the unrestricted one. The fourth shock does not disappear when its column is left free. In a square system three orthonormal columns determine the fourth up to sign, so the free column is whatever direction the other three leave over, and there is no a priori reason for it to resemble the risk premium shock. Both estimations decompose the same $\hat{\Sigma}$ on the same $T = 389$ windows, so every difference between them comes from the identifying content placed on the fourth column, not from the observables, the sample, or the restrictions on the other three.

First, do the restrictions do any work? Yes, and Table 13 says how much and where. Taken together the eight conditions retain 18.8 % of the RP-unrestricted set. Taken one at

a time, no condition removes more than about two fifths of the set of admissible rotations: individual pass rates range from 58.7 to 94.8 %.

Second, would the shock look similar without them? In its center, yes; in its precision, no. Table 14 puts the two versions of the fourth column of impact matrix B side by side. The unrestricted column still lowers equity prices¹² and the unpredictable component of the surprise, exactly as the baseline column does. The loading on the predictable component is close to zero in both. The magnitudes are attenuated but not qualitatively different, the unrestricted equity loading being about half of the baseline one. What the restrictions change is the width of the identified set. The 5-95% bands are up to twice as wide without them. The restrictions do not relocate the shock; they decide how sharply it is measured. Finally, removing every restriction on RP leaves the other three shocks almost unchanged: the correlation between each of the unrestricted system's first three shocks and its baseline counterpart is 0.95 or above. The identifying content placed on the fourth shock is not altering the first three.

Table 13: Share of the RP-unrestricted set that satisfies each RP restriction

Restriction on RP	Condition	Share (%)
Lowers the unpredictable part of the surprise	$a_{1,4} < 0$	58.7
Lowers the two-year yield	$a_{1,4} + a_{2,4} < 0$	62.8
Lowers equity prices	$a_{3,4} < 0$	64.0
Moves the ten-year more than MP does	$ a_{4,1} < a_{4,4} $	78.7
Loads more on the unpredictable part	$ a_{2,4} < a_{1,4} $	84.5
Moves against the risk-appetite index	$\text{corr}(\varepsilon_t^{RP}, \Delta BBM_t) < 0$	85.1
Moves the two-year less than MP does	$ a_{1,1} + a_{2,1} > a_{1,4} + a_{2,4} $	90.9
Moves the ten-year more than the two-year	$ a_{1,4} + a_{2,4} < a_{4,4} $	94.8
All eight jointly		18.8

Notes. Of 1 million rotations, 10,278 satisfy every restriction on MP, FRN and CBI. Each row reports the share of those 10,278 that also satisfies the stated RP restriction, taken one at a time; the last row reports the share satisfying all eight at once. Rows are ordered from most to least binding. The ten-year sign is the column normalization and is therefore not testable here.

¹²Absent the restrictions, 64.0 % of admissible rotations have a risk premium shock that lowers equity prices.

Table 14: Impact loadings of the RP column, with and without its restrictions

Row of B	Baseline	RP unrestricted
	median [5, 95]	median [5, 95]
RES	−0.011 [−0.024, −0.003]	−0.004 [−0.027, 0.019]
FIT	−0.001 [−0.005, 0.004]	−0.002 [−0.008, 0.004]
Δ S&P500	−0.160 [−0.426, −0.014]	−0.083 [−0.439, 0.282]
ΔR_{10Y}	−0.037 [−0.042, −0.031]	−0.032 [−0.043, −0.017]

Notes. Point estimates are the element-wise median across the identified set, not a single admissible rotation; brackets are the fifth and ninety-fifth %iles across the set. Units are %age points for RES, FIT and the ten-year yield, and % for the S&P 500, so magnitudes are comparable within a row but not across rows. The ten-year sign is imposed as the column normalization in both columns of the table.

9.4 Isolating the cost of an omitted shock

The three-shock systems in Section 5 differ from the baseline in two ways at once: they identify one shock fewer, and they impose only a subset of the restrictions the model in Section 2 provides. For example, the risk premium shock here must correlate negatively with the daily risk-appetite index of Bauer et al. (2023), which Cieslak–Schrimpf do not require, and the information shock must move RES more than FIT, which Jarocinski–Karadi do not. Thus, it is not clear whether the difference in results is due to the added shock or to the difference in restrictions.

This appendix separates the two by varying only the restrictions: each three-shock system is re-estimated under the restrictions of the four-shock system rather than its published ones. Because RES + FIT is the two-year surprise, both three-shock observable sets are nested in the baseline’s. Cieslak–Schrimpf give up the direction that separates FRN from MP, and Jarocinski–Karadi the direction that identifies RP. Restrictions that refer to what a system gives up cannot be imposed on it, but the rest carry over unchanged. Each re-estimated system therefore differs from the baseline in the missing shock alone, which is what makes the exercise possible.

Formally, write s_{mj} for the variance share defined in Section 5, with a superscript for the system: P for a three-shock system as published, P^* for the same system re-estimated under this paper’s restrictions, and JKCS for the baseline. For every observable m and every shock j the two systems share,

$$\underbrace{s_{mj}^{\text{JKCS}} - s_{mj}^P}_{\text{the gap in Section 5}} = \underbrace{s_{mj}^{P^*} - s_{mj}^P}_{\text{restrictions}} + \underbrace{s_{mj}^{\text{JKCS}} - s_{mj}^{P^*}}_{\text{missing shock}}. \quad (20)$$

Tables 15 and 16 report the results. Against Cieslak–Schrumpf, swapping the restrictions moves no share by more two %age points, while the shock omission moves them by up to thirteen: the gaps in Section 5 are therefore the price of the missing shock. Against Jarocinski–Karadi the opposite holds. The four-shock system adds several conditions on the shocks that system already has, including the one above, and they move the shares by more than the fourth shock does. Those gaps cannot be read as the cost of the omission alone.

Table 15: Variance-share gaps against Cieslak–Schrumpf, split into restrictions and the missing shock

Variable	MP	CBI	RA
<i>Restrictions</i>			
ΔR_{2Y}	+0.2	+0.3	-0.5
$\Delta \text{S\&P500}$	-0.6	-0.6	+1.2
ΔR_{10Y}	+1.7	-0.2	-1.4
<i>Missing shock</i>			
ΔR_{2Y}	-12.0	-7.5	-1.8
$\Delta \text{S\&P500}$	-13.3	+8.6	-3.3
ΔR_{10Y}	-9.0	+1.3	-2.9

Notes. The three-shock system is estimated twice on the same observables and with the same shocks, once under the restrictions Cieslak–Schrumpf impose and once under those restrictions of the four-shock system that its observables and shocks allow. Entries are announcement-window variance shares in %age points. Each system is evaluated at its own Fry–Pagan rotation, as in Section 5, and the two blocks add to the gap reported there.

Table 16: Variance-share gaps against Jarocinski–Karadi, split into restrictions and the missing shock

Variable	MP	FRN	CBI
<i>Restrictions</i>			
RES	-14.6	+0.0	+14.6
FIT	+0.3	+10.8	-11.1
Δ S&P500	+17.0	-9.0	-8.0
<i>Missing shock</i>			
RES	-2.9	+2.7	-2.5
FIT	+2.5	+6.0	-8.8
Δ S&P500	-0.1	-4.5	+0.7

Notes. The three-shock system is estimated twice on the same observables and with the same shocks, once under the restrictions Jarocinski–Karadi impose and once under those restrictions of the four-shock system that its observables and shocks allow. Entries are announcement-window variance shares in %age points. Each system is evaluated at its own Fry–Pagan rotation, as in Section 5, and the two blocks add to the gap reported there.

9.5 Daily responses of financial variables

Figures 12–15 report the daily local projections described in Section 6.1, one figure per shock and one panel per variable, with units given in each panel title. The sample keeps every announcement with the required daily data and the projection stops at 27 business days; the number of announcements retained differs across panels because the underlying series begin and end on different dates. The solid line is the specification without controls, the dashed line adds five lags of the daily change in the dependent variable, and the shaded band is the 5–95% range of the identified set combined with sampling uncertainty in the projection, for the specification without controls; the dotted lines trace the same range for the five-lag specification. The CDX series are total-return indices, so a *rising* line means credit is performing well, that is, risk on. BEX is the Bekaert–Engstrom–Xu index of risk *aversion* and therefore moves opposite to the Bauer et al. (2023) index of risk *appetite*.

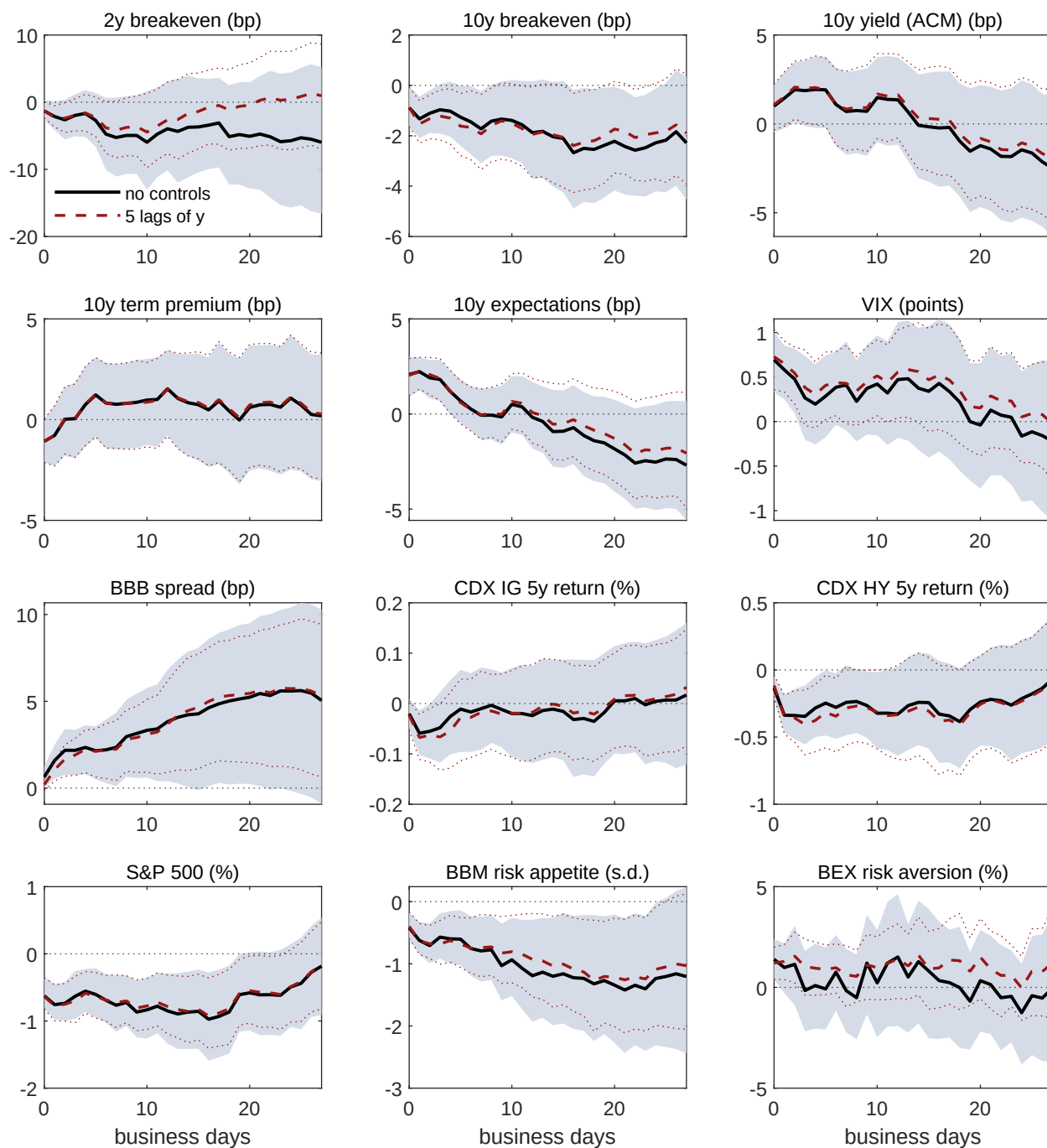


Figure 12: Daily responses to the monetary policy shock

Notes. Lines, bands and sample follow the description at the start of this appendix; units are given in each panel title.

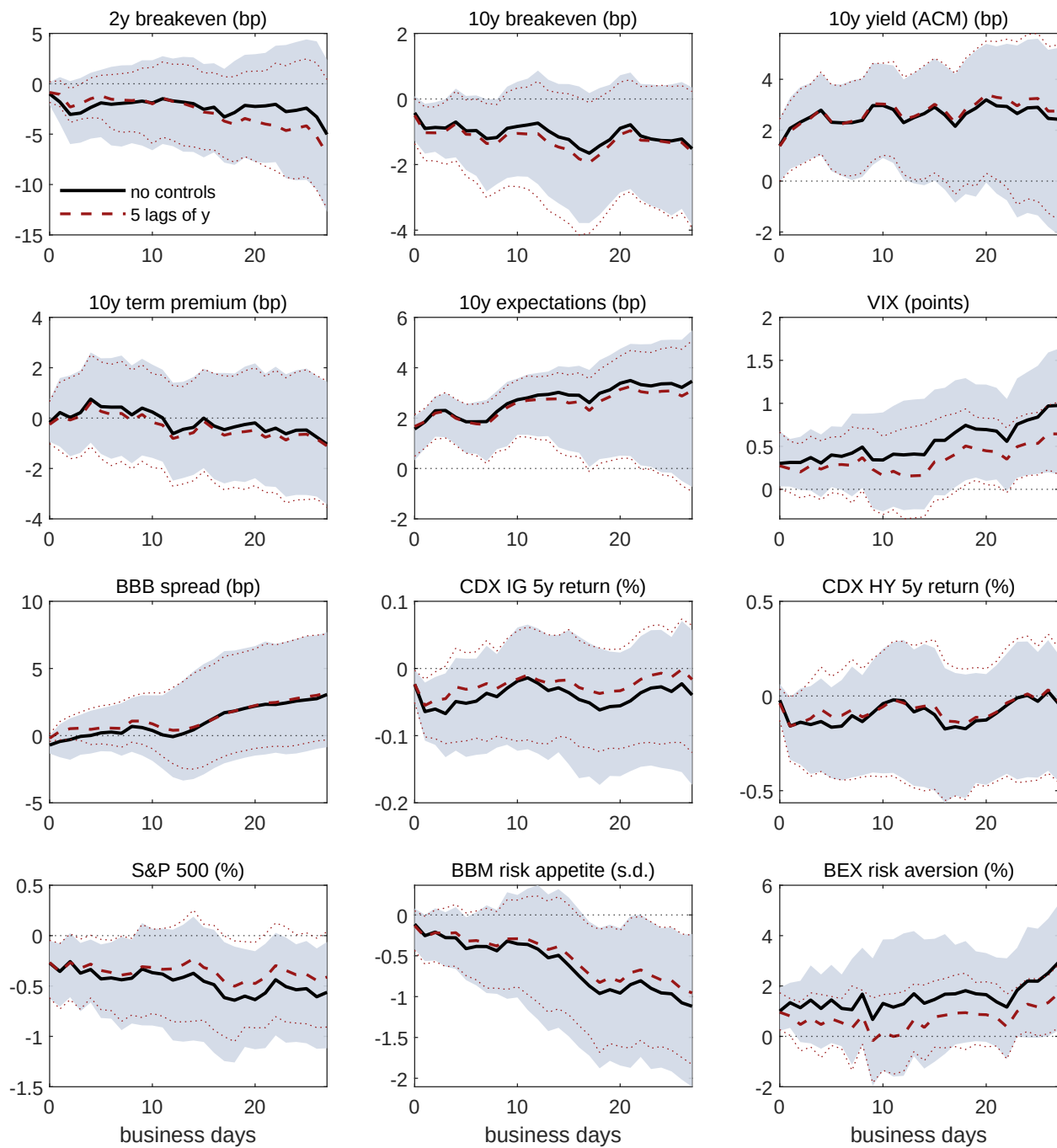


Figure 13: Daily responses to the Fed-response-to-news shock

Notes. Lines, bands and sample follow the description at the start of this appendix; units are given in each panel title.

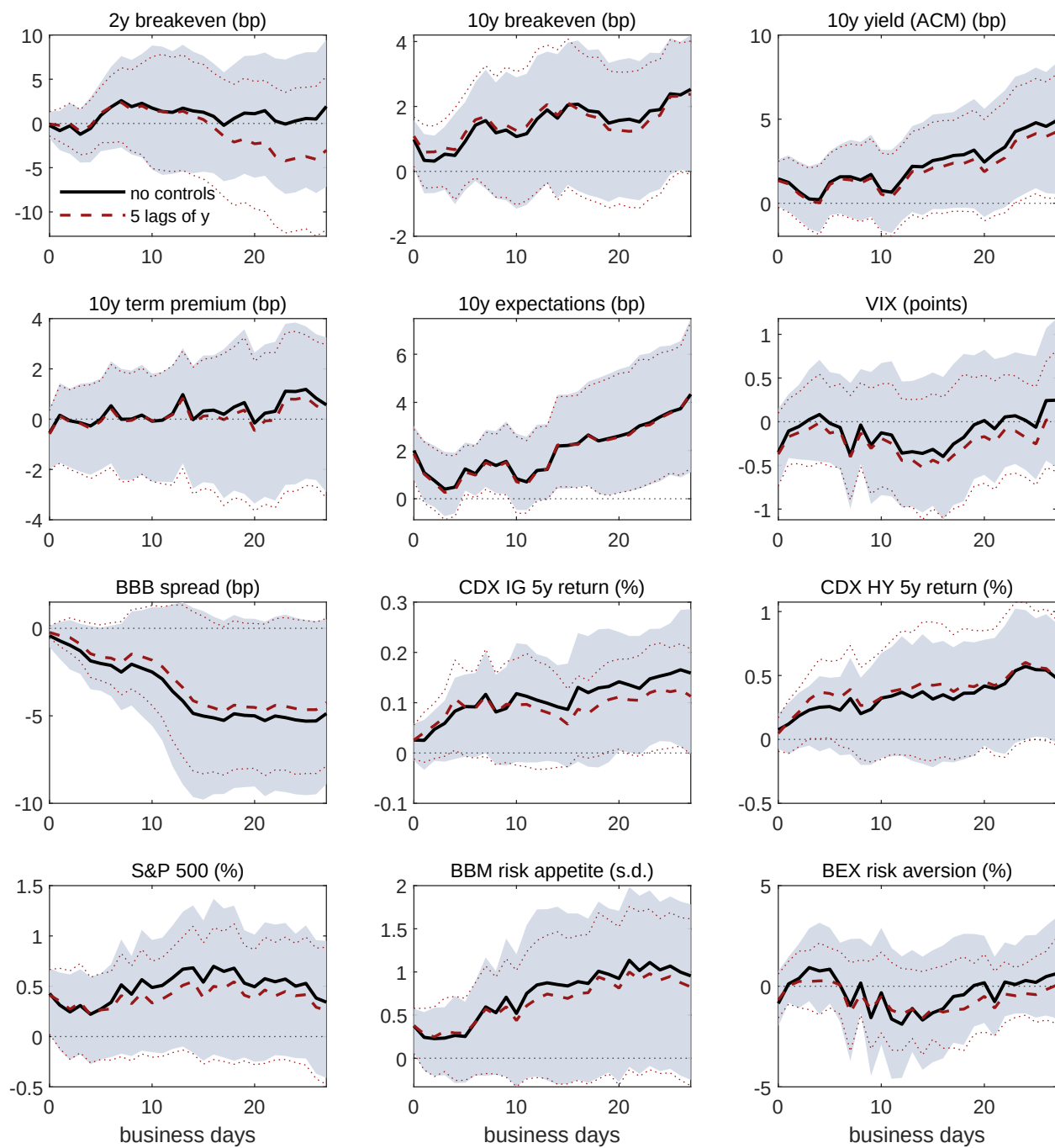


Figure 14: Daily responses to the central-bank information shock

Notes. Lines, bands and sample follow the description at the start of this appendix; units are given in each panel title.

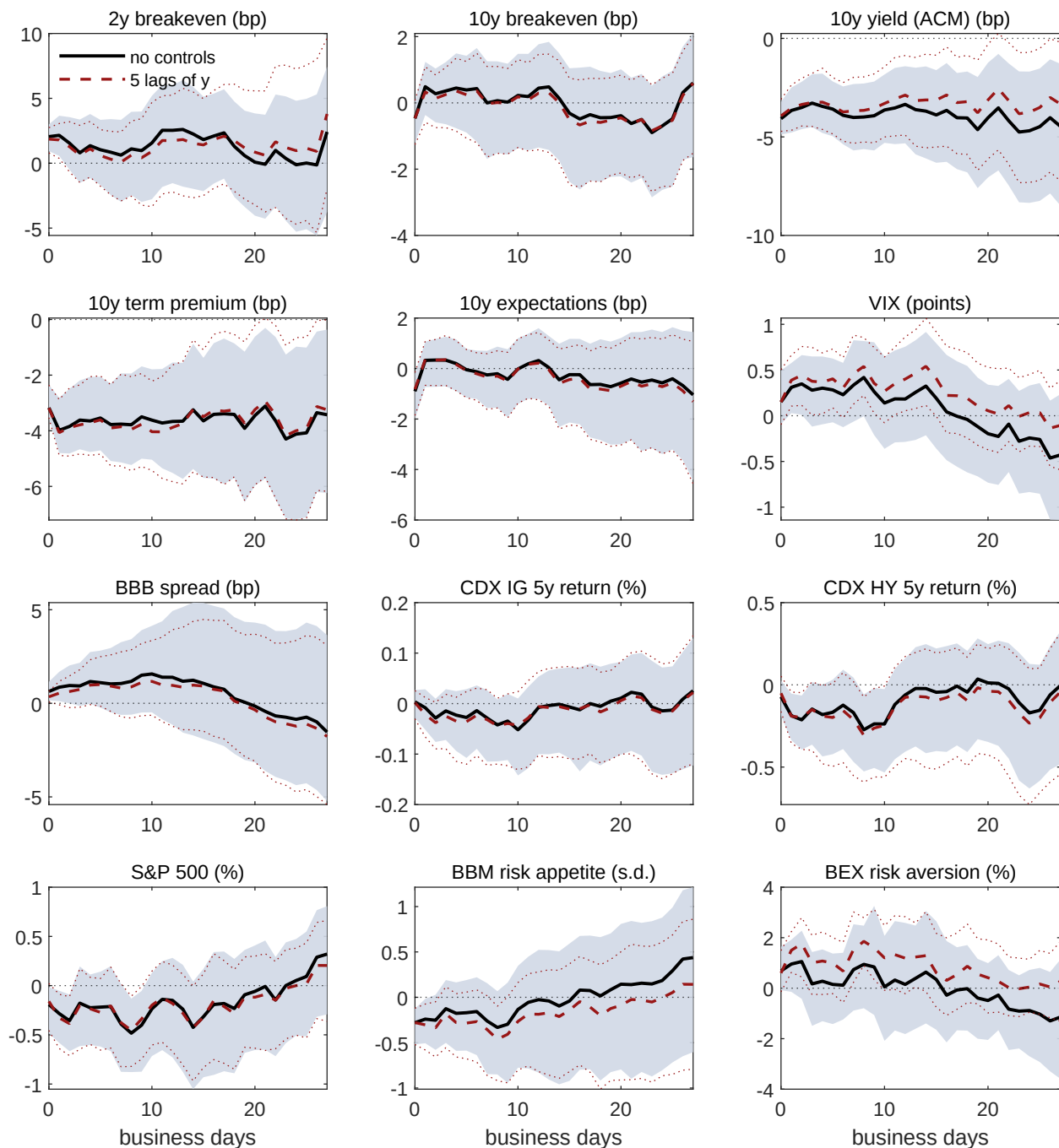


Figure 15: Daily responses to the risk-premium shock

Notes. Lines, bands and sample follow the description at the start of this appendix; units are given in each panel title.

What the projections look like with one shock missing

Figures 16 and 17 re-estimate Equation (17) on the shocks of the two three-shock systems of Section 5: the first identifies a policy, an information and a risk-premium shock but no

Fed-response-to-news shock, the second identifies a policy, a news and an information shock but no risk-premium shock. Everything else is held fixed, the same announcement days, the same sample rule, the same horizon and the same standard errors, so the only difference between the two lines in each panel is which system the shocks came from.

These comparisons are translations, not tests. Each three-shock system observes a strict subspace of the observables used here, so each of its shocks is an exact linear combination of the four shocks of this paper and its responses are an exact linear function of the responses in Figures 12–15. Neither figure can contradict them. The weights are close to the overlap shares of Table 7 but are not equal to them: those shares are computed over the full set of announcement windows, on which the four shocks are orthogonal by construction, whereas the projection is estimated on the restricted subset described in Section 6.1, on which they are not. What the figures add to the overlap table is units, what an omitted shock costs in basis points rather than in variance shares.

Omitting the Fed-response-to-news shock damages the policy shock. Figure 16 shows its information and risk-premium columns lying on top of their counterparts here, which is the visual check that the mapping works: those two shocks are 96 and 99 % their counterparts. The policy column is where the omission shows. By the end of the horizon the three-shock policy shock widens the BBB spread by about 6.4 basis points against 5.1 for the policy shock identified here, and lowers measured risk appetite by about 1.7 standard deviations against 1.2. The direction follows from the responses already reported: the news shock widens the spread by about 3 basis points on its own, and the three-shock policy shock loads on it with a weight of 0.45, the square root of its 19.9 % share, which accounts for the gap.

Omitting the risk-premium shock damages the news shock, and leaves the other two close to their four-shock values. In Figure 17 the three-shock policy and information shocks reproduce the equity responses reported here to within 0.2 %, and the credit and risk-appetite responses with end-of-horizon gaps less than half the size of the news-shock gaps, while the news shock widens the BBB spread by about 4.7 basis points against 3.1 and lowers risk appetite by about 1.5 standard deviations against 1.1, overstatements of about a half and a third. The asymmetry is the same one documented in Section 5 and it runs in the direction that section predicts.

Both figures compare point estimates. Every three-shock response lies inside the band of its counterpart, so they support statements about the magnitude a three-shock reading would attribute to a given label, and not statements that the systems are statistically distinguishable at daily frequency.

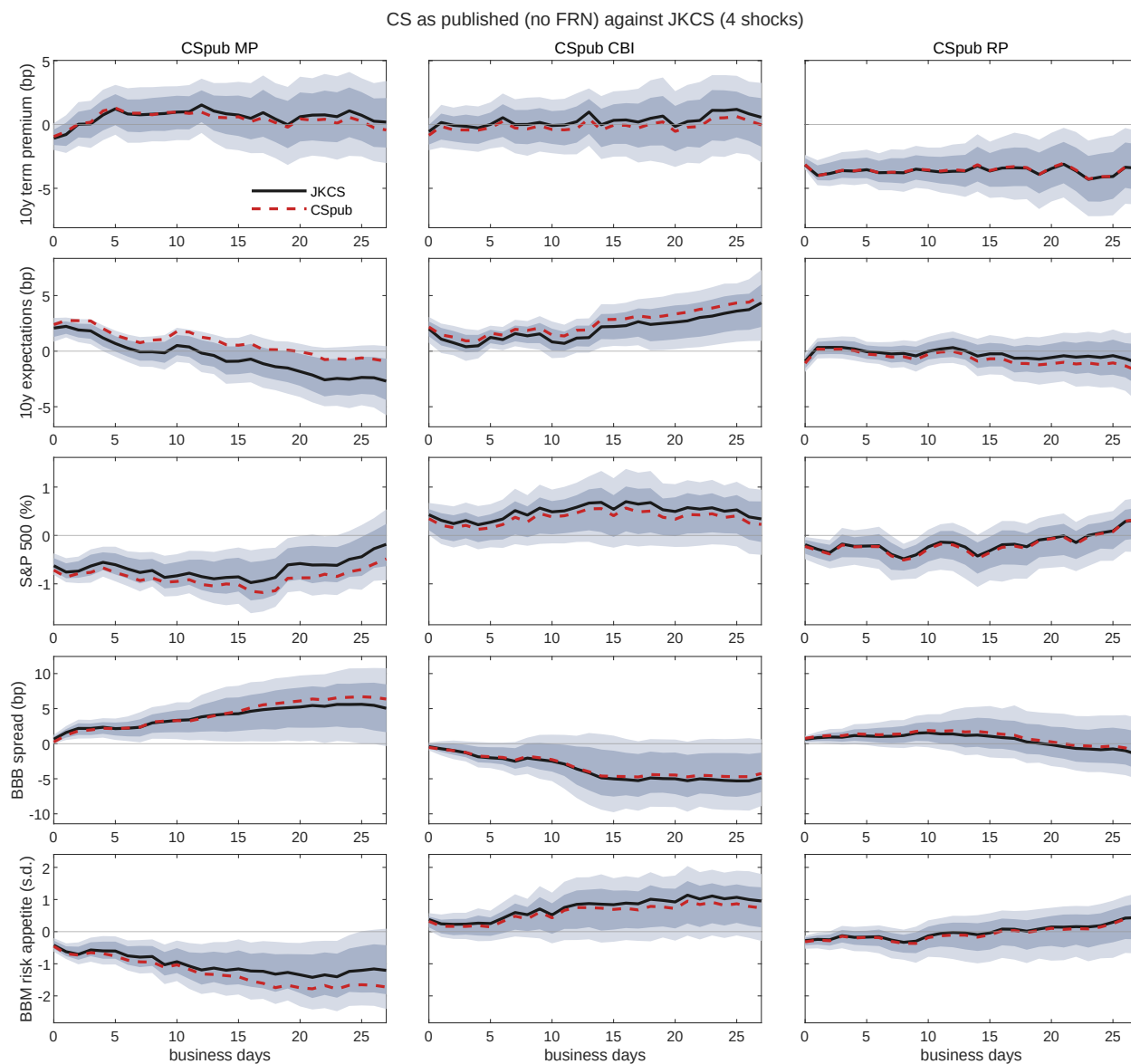


Figure 16: Daily responses with and without a Fed-response-to-news shock

Notes. Rows are variables and columns are the three shocks of the three-shock system; the share of each shock that is its counterpart in this paper is reported in Table 7. The solid line and the shaded bands are the four-shock system of Figure 7; the dashed line is the three-shock system. Both are estimated on the same announcement days with the same specification.

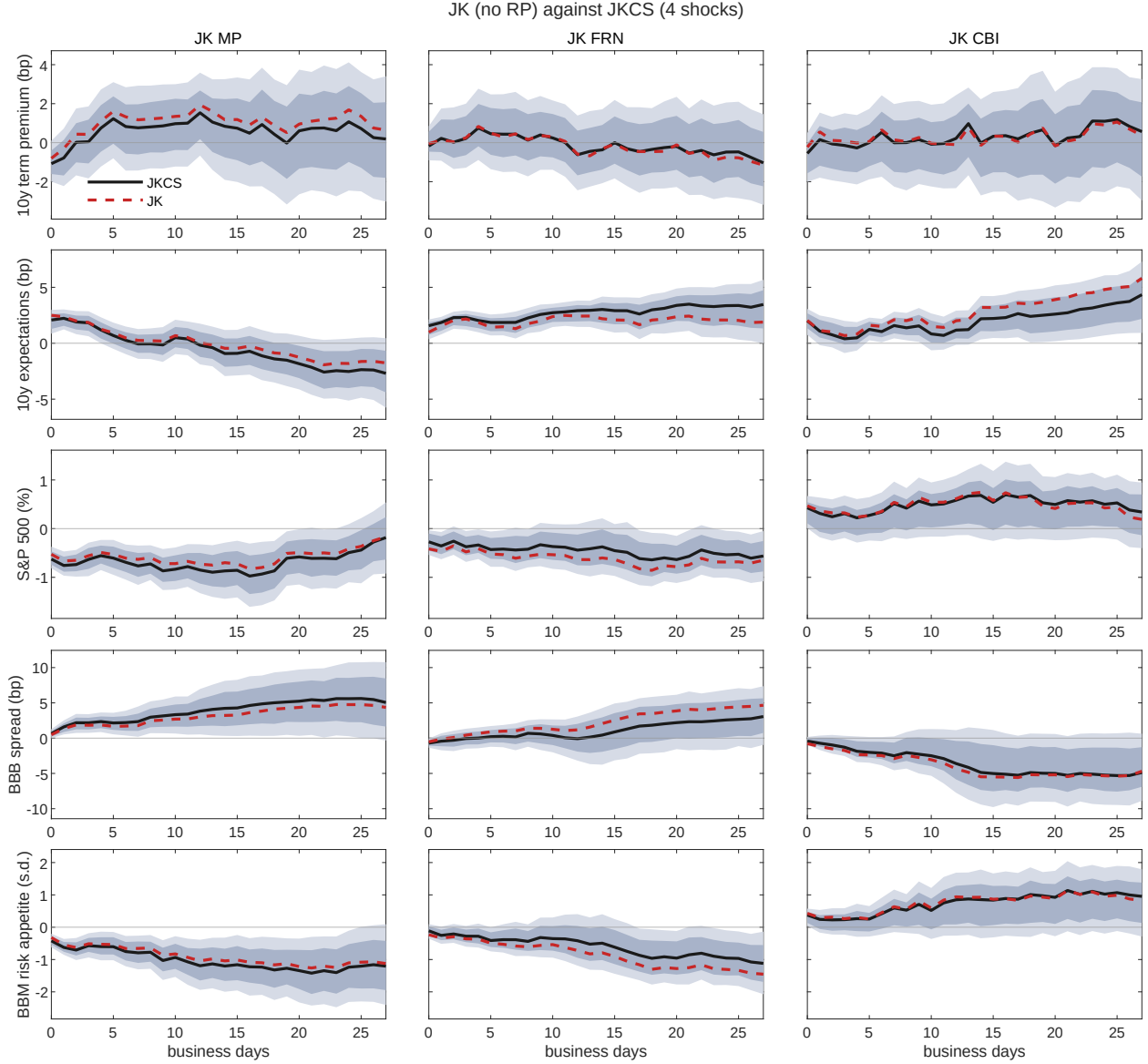


Figure 17: Daily responses with and without a risk-premium shock

Notes. Layout as in Figure 16.

9.6 Model appendix: bond loadings

This appendix derives the three yield loadings B_n^i , B_n^g , B_n^x used in Section 2.4 and establishes the two shape facts the identification relies on: $B_1^x = 0$, so the risk premium shock does not move the one-period yield, and $|B_n^x|$ keeps rising further out the curve than B_n^g does, so the premium loading is the one that dominates at the long end.

Let $P_t^{(n)}$ be the price of an n -period zero-coupon bond and $p_t^{(n)} = \log P_t^{(n)}$. Since $y_t^{(n)} =$

$-\frac{1}{n}p_t^{(n)}$, write the log price as

$$p_t^{(n)} = -\bar{A}_n - \bar{B}'_n F_t, \quad \text{so} \quad y_t^{(n)} = \frac{\bar{A}_n}{n} + \frac{\bar{B}'_n}{n} F_t \equiv A_n + B'_n F_t.$$

Holding the n -bond for one period gives $P_t^{(n)} = \mathbb{E}_t[M_{t+1} P_{t+1}^{(n-1)}]$, so $p_t^{(n)} = \log \mathbb{E}_t[e^Z]$ with $Z \equiv m_{t+1} + p_{t+1}^{(n-1)}$. Substituting the SDF and $F_{t+1} = \Phi F_t + \Sigma \varepsilon_{t+1}$,

$$Z = \underbrace{-i_t - \frac{1}{2}\lambda'_t \lambda_t - \bar{A}_{n-1} - \bar{B}'_{n-1} \Phi F_t}_{\mathbb{E}_t[Z]} - \underbrace{(\lambda_t + \Sigma' \bar{B}_{n-1})' \varepsilon_{t+1}}_{\text{mean-zero given } t}.$$

Z is Gaussian given t , so $\mathbb{E}_t[e^Z] = e^{\mathbb{E}_t[Z] + \frac{1}{2} \text{Var}_t(Z)}$ with $\text{Var}_t(Z) = v'v$ and $v \equiv \lambda_t + \Sigma' \bar{B}_{n-1}$. Expanding $v'v$, the $\pm \frac{1}{2}\lambda'_t \lambda_t$ terms cancel and one cross term survives:

$$\lambda'_t \Sigma' \bar{B}_{n-1} = \lambda_{gx} \sigma_g D_{n-1} x_t, \quad D_n \equiv \bar{B}_n^g + a^M \bar{B}_n^i.$$

This is linear in F_t through $x_t = e'_x F_t$, so it enters the loading and not the constant. Matching against the guess $p_t^{(n)} = -\bar{A}_n - \bar{B}'_n F_t$,

$$\bar{A}_n = \bar{A}_{n-1} - \frac{1}{2} \bar{B}'_{n-1} \Sigma \Sigma' \bar{B}_{n-1}, \quad \bar{B}_n = e_i + \Phi' \bar{B}_{n-1} - \lambda_{gx} \sigma_g D_{n-1} e_x.$$

Writing out the three components of $\Phi' \bar{B}_{n-1}$ gives

$$\bar{B}_n^g = \phi_g (\bar{B}_{n-1}^g + a^M \bar{B}_{n-1}^i), \quad \bar{B}_n^i = 1 + \phi_i \bar{B}_{n-1}^i, \quad \bar{B}_n^x = \phi_x \bar{B}_{n-1}^x - \lambda_{gx} \sigma_g D_{n-1}.$$

The starting values come from the one-period bond, which pays 1 and so has $P_t^{(1)} = \mathbb{E}_t[M_{t+1}] = e^{-i_t}$: hence $\bar{A}_1 = 0$ and $\bar{B}_1 = e_i$, that is $\bar{B}_1^i = 1$ and $\bar{B}_1^g = \bar{B}_1^x = 0$. The one-period yield *is* the short rate, and it carries no premium. With $S_n(\phi) \equiv 1 + \phi + \dots + \phi^{n-1} = \frac{1-\phi^n}{1-\phi}$ the three recursions solve to

$$B_n^i = \frac{S_n(\phi_i)}{n} > 0, \quad B_n^g = \frac{a^M \phi_g}{n} \frac{S_n(\phi_g) - S_n(\phi_i)}{\phi_g - \phi_i} > 0, \quad B_n^x = -\frac{\lambda_{gx} \sigma_g}{n} \sum_{k=1}^{n-1} \phi_x^{n-1-k} D_k \leq 0,$$

with $D_k \equiv \bar{B}_k^g + a^M S_k(\phi_i) > 0$ and equality in the last expression only at $n = 1$, where the sum is empty.

Shapes. B_n^i is decreasing in n . B_n^g rises from $B_1^g = 0$ to an interior peak and then declines toward zero. So does $|B_n^x|$, and two induction steps show why it peaks later. First, $\bar{B}_n^g$ is increasing, since $\bar{B}_n^g - \bar{B}_{n-1}^g = \phi_g (\bar{B}_{n-1}^g - \bar{B}_{n-2}^g) + a^M \phi_g (\bar{B}_{n-1}^i - \bar{B}_{n-2}^i) > 0$ from the base

step $\bar{B}_2^g - \bar{B}_1^g = a^M \phi_g > 0$, so D_k is increasing. Second, writing $q_n \equiv -\bar{B}_n^x \geq 0$, the recursion $q_n = \phi_x q_{n-1} + \lambda_{gx} \sigma_g D_{n-1}$ with $q_1 = 0$ gives $q_n - q_{n-1} > 0$ by the same argument, so $|\bar{B}_n^x|$ is strictly increasing in n . Since D_k converges to $D_\infty = a^M / [(1 - \phi_g)(1 - \phi_i)]$, the log-price loading $\bar{B}_n^x$ grows asymptotically linearly and the yield loading $B_n^x = \bar{B}_n^x / n$ therefore returns to zero, attaining an interior peak far out in n when x is near a unit root.

Two consequences for the identification. The peak of $|B_n^x|$ always lies beyond the peak of B_n^g in n , and $|B_n^x|$ reaches the peak farther in n with higher ϕ_x , given all other things equal. For example, at $\phi_g = .85$ and $\phi_i = .95$ and near unit root process for x , $\phi_x = 0.99$, the peak of $|B_n^x|$ happens at $n = 91$, being roughly 11 years given eight announcements per year. This delivers $|B_{N_{10}}^x| > |B_{N_2}^x|$ restriction, meaning that risk premium shock affects longer term rates more than shorter term rates.

The two slope entries of Table 1 follow. Writing N_2 and N_{10} for the two- and ten-year maturities, the CBI entry is $a(B_{N_2}^i - B_{N_{10}}^i) > B_{N_{10}}^g - B_{N_2}^g$, whose left side is always positive and whose right side is negative whenever the hump in B_n^g peaks inside two years, in which case the inequality holds automatically. It can fail only when ϕ_g and ϕ_i are both near one. The RP entry is $|B_{N_{10}}^x| > |B_{N_2}^x|$, which holds when x is persistent and fails when x mean-reverts quickly.

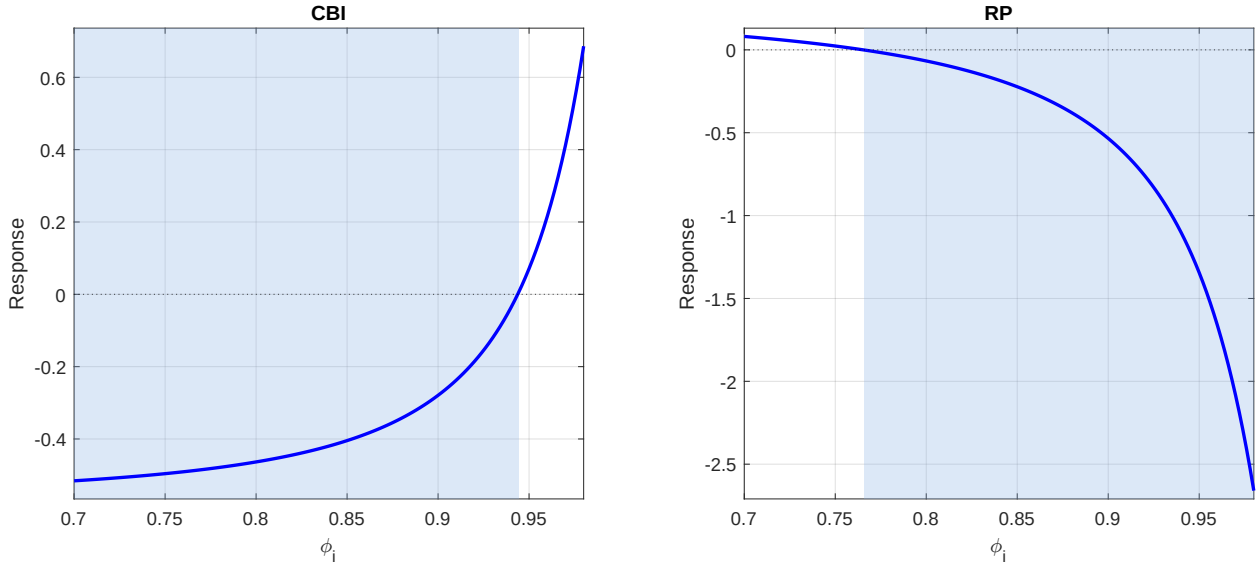


Figure 18: Admissible policy-rate persistence

Notes. The slope response (the ten-year minus the two-year response, scaled by the magnitude of the two-year response) is plotted against the policy-rate persistence ϕ_i , with the growth and price-of-risk persistences held at their calibrated values. Shaded areas indicate the region where both $\dagger$ conditions hold. RP sets the lower bound and CBI the upper, leaving $\phi_i \in [0.766, 0.945]$, equivalently $[0.59, 0.89]$ in quarterly terms.

9.6.1 Expected rates and the term premium

This subsection shows that the split of the yield response into an expected-rates part and a term premium is the split of B_n into (B_n^i, B_n^g) part and B_n^x part, so that the risk premium shock moves the premium and nothing else.

Iterate the policy rule forward under the market's beliefs. Expected growth is $\mathbb{E}_t^M[g_{t+j}] = \phi_g^j g_t$, and the rule gives $\mathbb{E}_t^M[i_{t+j}] = \phi_i \mathbb{E}_t^M[i_{t+j-1}] + a^M \phi_g^j g_t$ with $\mathbb{E}_t^M[i_t] = i_t$. Neither the rule nor the growth process contains x , so the risk factor never enters: expected future short rates depend on i_t and g_t alone. Unrolling from $j = 0$,

$$\mathbb{E}_t^M[i_{t+j}] = \phi_i^j i_t + a^M \phi_g \frac{\phi_g^j - \phi_i^j}{\phi_g - \phi_i} g_t,$$

and averaging over $j = 0, \dots, n-1$ with $\sum_{j=0}^{n-1} \phi^j = S_n(\phi)$ returns exactly the loadings of Appendix 9.6,

$$\frac{1}{n} \sum_{j=0}^{n-1} \mathbb{E}_t^M[i_{t+j}] = B_n^i i_t + B_n^g g_t, \quad \text{so} \quad TP_t^{(n)} = y_t^{(n)} - (B_n^i i_t + B_n^g g_t) = A_n + B_n^x x_t,$$

which loads on x alone. Differencing over the announcement window, with $\Delta g_t = \sigma_g \varepsilon_t^g$, $\Delta x_t = \sigma_x \varepsilon_t^x$ and A_n constant, gives the responses reported in Section 2.4: MP, CBI and FRN move only g and i , and therefore only expected rates.

9.7 Model appendix: stocks loadings

This appendix solves the stock block and establishes three things: the price loadings B_s used in Section 2.4; that the risk premium shock is the only shock that moves the equity premium; and that it moves the equity premium and the term premium in *opposite* directions, which is what produces the stocks-down, yields-down signature the identification restricts.

Setup. Log dividend growth has conditional mean g_t and an independent cash-flow shock, $\Delta d_{t+1} = g_t + \sigma_d \varepsilon_{t+1}^d$ with $\varepsilon^d \perp (\varepsilon^g, \varepsilon^i, \varepsilon^x)$. The Campbell–Shiller identity, log-linearized around the mean log price–dividend ratio $\overline{pd}$, is $r_{s,t+1} = \kappa_0 + \kappa_1 pd_{t+1} - pd_t + \Delta d_{t+1}$ with $\kappa_1 = e^{\overline{pd}} / (1 + e^{\overline{pd}}) \in (0, 1)$. This is the one approximation in the model; the bond block is exact. Guess $pd_t = A_s + B_s' F_t$.

Solution. Pricing is under the market's beliefs, so Φ and Σ are evaluated at a^M . The stock Euler equation $1 = \mathbb{E}_t[e^{m_{t+1} + r_{s,t+1}}]$ with a conditionally Gaussian exponent is $0 =$

$\mathbb{E}_t[m_{t+1} + r_{s,t+1}] + \frac{1}{2} \text{Var}_t[m_{t+1} + r_{s,t+1}]$. Collecting the terms in F_t and setting their coefficient to zero gives

$$B_s = \kappa_1 \Phi' B_s + e_g - e_i - \kappa_1 \lambda_{gx} \sigma_g D_s e_x, \quad D_s \equiv B_s^g + a^M B_s^i,$$

where $-e_i$ comes from the SDF, e_g from dividends, and the last term from the cross moment $-\kappa_1 \lambda'_t \Sigma' B_s = -\kappa_1 \lambda_{gx} \sigma_g D_s x_t$ in the variance. Component by component, using $\Phi' B_s = (\phi_g D_s, \phi_i B_s^i, \phi_x B_s^x)'$, the three equations solve to

$$B_s^i = \frac{-1}{1 - \kappa_1 \phi_i} < 0, \quad D_s = \frac{1 + a^M B_s^i}{1 - \kappa_1 \phi_g}, \quad B_s^g = \kappa_1 \phi_g D_s + 1, \quad B_s^x = \frac{-\kappa_1 \lambda_{gx} \sigma_g D_s}{1 - \kappa_1 \phi_x}$$

with $B_s^i < -1$, since $0 < \kappa_1 \phi_i < 1$.

The equity premium. Substituting $-i_t = \mathbb{E}_t[m_{t+1}] + \frac{1}{2} \text{Var}_t(m_{t+1})$ into the Euler equation collapses every term involving m alone, and what remains defines and evaluates the premium:

$$EP_t \equiv \mathbb{E}_t[r_{s,t+1}] - i_t + \frac{1}{2} \text{Var}_t(r_{s,t+1}) = -\text{Cov}_t(m_{t+1}, r_{s,t+1}),$$

the expected excess log return plus the Jensen term. The return innovation is $(\kappa_1 \Sigma' B_s)' \varepsilon_{t+1} + \sigma_d \varepsilon_{t+1}^d$, so its loading on the priced shock is $\beta_g = \kappa_1 \sigma_g D_s$; since the SDF innovation is $-\lambda_{gx} x_t \varepsilon_{t+1}^g$ and the shocks are orthogonal with unit variance, every other term drops and

$$EP_t = \lambda_{gx} x_t \beta_g = \kappa_1 \lambda_{gx} \sigma_g D_s x_t, \quad \Delta EP_t = \kappa_1 \lambda_{gx} \sigma_g D_s \sigma_x \varepsilon_t^x, \quad \Delta EP_t|_{\text{MP, CBI, FRN}} = 0.$$

Three things follow. First, EP_t is proportional to x_t and to nothing else, so RP is the only shock that moves it. Second, the two premia move in opposite directions on that shock: $\Delta TP_t^{(n)} < 0$ while $\Delta EP_t > 0$. A positive RP shock raises the compensation demanded on equity, which is exposed to growth risk, and lowers it on bonds, which hedge growth risk. Stock prices fall while bond prices rise, the flight-to-quality co-movement the identification restricts. Third, the premium and the price loading on x are one object seen twice: substituting for B_s^x gives $EP_t = -(1 - \kappa_1 \phi_x) B_s^x x_t$, so a unit rise in x raises the required return by $(1 - \kappa_1 \phi_x) |B_s^x|$ per period and lowers the log price by $|B_s^x|$, the capitalized value of those higher required returns.

Window response and signs. Within the window dividends are predetermined ($\varepsilon_t^d = 0$), so the log-price change is the revaluation of pd , $\Delta \log P_t = B_s' \Delta F$ with $\Delta F = (\sigma_g \varepsilon_t^g, \Delta y_t^{(1)}, \sigma_x \varepsilon_t^x)'$, which delivers the four responses of Section 2.4. Since $B_s^i < 0$, MP and FRN lower the price

unconditionally. The other two signs are conditional, on different conditions. RP lowers the price iff $B_s^x < 0$, that is iff

$$D_s > 0 \quad \Longleftrightarrow \quad a^M < 1 - \kappa_1 \phi_i,$$

which is the same condition that makes the unconditional equity premium positive: a positive equity premium and the RP sign restriction are the same regime. CBI raises the price iff the cash-flow channel dominates the discount channel, $B_s^g + a B_s^i > 0$, which after substitution and multiplication by $(1 - \kappa_1 \phi_g)(1 - \kappa_1 \phi_i) > 0$ is

$$(1 - \kappa_1 \phi_g) a + \kappa_1 \phi_g a^M < 1 - \kappa_1 \phi_i,$$

the same bound imposed on a weighted average of the true and the perceived growth response. Since $a > a^M$, the left side exceeds a^M , so the CBI condition is strictly tighter than the RP condition and implies it. This is why the CBI entry in Table 1 is the binding one.